IA321 AI Project Final Report: Audio Deepfake Detectability Depends Strongly on Generator Family

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***Abstract*—Recent progress in generative audio has made synthetic speech and general audio increasingly realistic, but it has also made deepfake-audio detection more heterogeneous. In this work, we study how the *generator family* shapes downstream detectability. We build reproducible synthetic-audio datasets from two distinct generation settings: voice conversion (VC), using DDDM-VC and Diff-HierVC, and text-to-audio (TTA), using AudioLDM and AudioLDM 2 [1]–[4]. We then evaluate three detector families with increasing complexity: a lightweight MFCC-SVM baseline, spectrogram-based CNNs, and the stronger AASIST3 anti-spoofing architecture [5], [6].**

Our main finding is not simply that some detectors outperform others, but that *VC and TTA induce different detection regimes*. Across detectors, VC deepfakes leave much stronger and more consistent traces than TTA deepfakes. AASIST3 reaches near-perfect performance on the VC dataset, whereas performance degrades substantially on TTA. Likewise, the MFCC-SVM baseline is nearly perfect on VC but markedly weaker on TTA, and spectrogram CNNs show poor cross-domain transfer. Explainability analysis on the SVM using SHAP, LIME, and variance-based Sobol indices further clarifies this gap: VC detection is dominated by a small number of strong cepstral cues, whereas TTA detection depends on weaker and more distributed statistics [7]–[9].

Taken together, these results suggest that audio deepfake detection should not be treated as a single homogeneous problem. Instead, robustness depends strongly on the generator family, the target domain, and the representation on which the detector operates. Our datasets and code are released to support reproducible follow-up work on generation, detection, and explainability.

I. Introduction

A. Context and Motivation

Deepfake audio has become a fast-moving research area at the intersection of generative modeling, speech processing, and trustworthy AI. On the generation side, recent years have seen major advances in zero-shot voice conversion, expressive text-to-speech, and text-to-audio generation, driven in part by diffusion models and large self-supervised representations [1]–[4], [10], [11]. On the detection side, the field has moved from handcrafted cepstral front-ends toward stronger end-to-end and self-supervised anti-spoofing systems such as RawNet2 and AASIST-style architectures [5], [6], [12]. At the same time, explainability has become increasingly important, because strong accuracy alone does not guarantee that detectors rely on meaningful synthesis artifacts rather than on spurious dataset shortcuts [13]–[15].

This evolution creates an important practical tension. Modern generative systems do not all fail in the same way, and synthetic audio is no longer limited to a single modality or use case. Voice conversion aims to preserve linguistic content while transferring speaker identity [10], [16]. Text-to-audio instead generates a broader class of audio signals from free-form captions, often with more diverse acoustic scenes and weaker structural constraints [3], [17], [18]. Text-to-speech occupies yet another regime, emphasizing intelligibility, alignment, prosody, and speaker control [11], [19]. From a detection perspective, these families should not be expected to leave the same traces. A detector that works extremely well on one synthetic-audio family may generalize poorly to another, even if both are described under the umbrella term *deepfake audio*.

This generator-dependence is central to the present report. Rather than treating deepfake-audio detection as a single closed problem, we study it as a pipeline linking *generation*, *detection*, and *explainability*. The generation stage determines not only the realism of the synthesized samples, but also the kind of residual artifacts that remain available to downstream detectors. The detection stage then tests whether those artifacts are strong, weak, localized, or transferable across domains. Finally, explainability helps determine whether the detector is capturing synthesis-specific cues or only exploiting superficial correlations [7]–[9], [20].

B. Project Objectives

The first objective of this project is to build a controlled and reproducible benchmark spanning multiple generator families. Concretely, we construct two synthetic-audio datasets: one based on *voice conversion* using DDDM-VC and Diff-HierVC, and one based on *text-to-audio* generation using AudioLDM and AudioLDM 2 [1]–[4]. We also review and partially implement a text-to-speech branch, although it is not completed at sufficient scale for inclusion in the final benchmark. A key design goal throughout is reproducibility: deterministic manifests, fixed splits, documented preprocessing, and traceable provenance for generated outputs.

The second objective is to compare detectors across levels of complexity. We intentionally do not restrict the study to a single backbone. Instead, we evaluate three detector families: a lightweight linear SVM trained on MFCC statistics, spectrogram CNN baselines based on VGG16 and ResNet18, and the more specialized AASIST3 anti-spoofing model [6], [21]–[23]. This design lets us distinguish what can already be captured by simple acoustic statistics from what requires more expressive convolutional or graph-based reasoning.

The third objective is to study explainability in relation to detectability. We do not treat explainability as an isolated add-on, but as a way to interpret the differences observed between generator families. In particular, we analyze the SVM detectors with SHAP, LIME, and variance-based Sobol attribution in order to identify whether VC and TTA produce different kinds of discriminative cues [7]–[9]. This is especially relevant because recent literature has emphasized an explainability gap in audio deepfake detection, particularly under realistic and cross-domain settings [14], [15].

The central research question is therefore the following: *do different audio-generation families induce different downstream detection regimes, and can explainability help characterize that difference?* Our results indicate that the answer is yes. Across detectors, VC deepfakes are consistently easier to detect than TTA deepfakes, and the explainability analysis suggests that this gap is tied to the concentration versus dispersion of the underlying acoustic cues.

Code and data for this project are released for reproducibility:

- **VC-Dataset repository:** https://huggingface.co/datasets/amine-maazizi/vc-detection-10k
- **TTA-Dataset repository:** https://huggingface.co/datasets/amine-maazizi/tta-detection-2k
- **Code repository:** https://github.com/amine-maazizi/diffusion-audio-deepfake-generation-detection/tree/main

C. Report Organization

The remainder of the report is structured as follows. Section II reviews prior work on deepfake-audio generation, detection, and explainability. Section III-A presents the generation models and dataset-construction pipeline used in our experiments. The following detection section introduces the SVM, CNN, and AASIST3 detectors and reports their performance across datasets and settings. The explainability section then studies which features drive the SVM decisions and how these explanations differ between VC and TTA. Finally, the last section synthesizes the main findings, discusses their limitations, and outlines directions for future work in broader generator coverage, more faithful interpretability, and multimodal detection.

II. Related Works

This section provides a high-level overview of the lines of work most relevant to our project. We first review deepfake audio generation, focusing on voice conversion, Text-to-audio, and Text-to-speech systems, since these models define the synthetic data considered in our experiments. We then summarize the evolution of deepfake audio detection from early feature-based approaches to modern self-supervised and graph-based detectors, and briefly position AASIST3 within this landscape. Finally, we introduce the main families of explainability methods used for audio and multimodal models. In all cases, we intentionally keep the discussion concise here and defer architectural and implementation details to the following sections.

A. Deepfake Audio Generation

Voice Conversion. Voice conversion (VC) aims to modify a source utterance so that it preserves the linguistic content of the source speaker while matching the identity of a target speaker. Early zero-shot VC systems relied on bottlenecked autoencoders or explicit prosodic conditioning to encourage speaker–content disentanglement, as in AutoVC and F0-AutoVC [16], [24]. Later approaches leveraged self-supervised speech representations and perturbation-based disentanglement to improve generalization to unseen speakers and better preserve phonetic content [25], [26]. More recent diffusion-based methods, such as DiffVC, DDDM-VC, and Diff-HierVC, further improved naturalness and speaker similarity by replacing deterministic decoding with iterative denoising and richer conditioning strategies [1], [2], [10]. Since VC is one of the generation settings used in our experiments, we describe the corresponding models in more detail in Section III-A.

Text-to-audio. Text-to-audio (TTA) generation aims to synthesize a general audio signal from a natural-language description. Earlier systems such as DiffSound and AudioGen generated compressed audio representations directly from text, but remained constrained by the limited scale and quality of paired audio–text data [17], [18]. Latent-diffusion approaches then improved scalability by operating in compressed spectrogram spaces rather than directly on waveform tokens [3], [27]. In particular, AudioLDM introduced a strong latent-diffusion framework based on a pretrained CLAP space, while AudioLDM 2 extended this idea through a two-stage semantic-to-acoustic pipeline based on the so-called language of audio [3], [4]. These two models are the main TTA baselines used in our work and are described more precisely in Section III-A.

Text-to-speech. Text-to-speech (TTS) generation focuses on synthesizing intelligible and natural speech from text, typically while controlling speaker identity, prosody, or speaking style. Flow-based models such as Glow-TTS showed that high-quality and parallelizable speech synthesis could be achieved through monotonic alignment and invertible generative modeling [19]. More recent systems such as HierSpeech++ combine hierarchical generation, self-supervised speech representations, explicit pitch modeling, and voice prompting to improve zero-shot speech synthesis and voice conversion [11]. In the context of our project, TTS provides a third family of fake-audio generators complementary to VC and TTA. We therefore only discuss these models here at a high level and defer their detailed description to Section III-A.

B. Deepfake Audio Detection

Deepfake audio detection has progressed rapidly alongside advances in speech generation. As Text-to-speech and voice-conversion systems become increasingly realistic, synthetic speech poses growing risks to speaker verification, media integrity, and forensic analysis. The literature has therefore evolved from handcrafted acoustic features toward end-to-end deep models and, more recently, toward self-supervised and graph-based detectors [28], [29]. In this section, we only summarize this trajectory at a high level and defer the detailed presentation of our detector to the detection section.

From Handcrafted Features to End-to-End Detection. Early anti-spoofing systems relied on handcrafted descriptors such as MFCCs, LPCCs, and residual-based cepstral features, motivated by the assumption that synthetic speech introduces artifacts in the excitation source, spectral envelope, or phase structure [28]. As spoofing methods improved, these fixed front-ends became less effective, which motivated a transition toward end-to-end neural detectors. Models such as RawNet2 illustrated this shift by learning waveform-level front-ends directly from data rather than relying on fixed spectrogram pipelines [12].

The AASIST Family and Graph-Based Modeling. A major step in this evolution was the introduction of the AASIST family (Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks), which models interactions across temporal and spectral regions through graph attention [5]. This design is particularly relevant for anti-spoofing because synthetic artifacts are often sparse and localized rather than globally distributed. Later variants such as AASIST-L explored lighter versions of the same idea for improved efficiency and deployment [30].

AASIST3 and SSL-Based Detection. Recent work increasingly combines graph-based backbones with large pretrained self-supervised speech encoders. AASIST3 follows this direction by integrating stronger SSL front-ends together with more flexible nonlinear transformations, and has emerged as one of the strongest recent anti-spoofing backbones, especially in ASVspoof-inspired settings [6], [31]. Since AASIST3 is also the basis of our own detector, we only position it here and present its architecture more precisely in the detection section.

Self-Supervised Learning as a Dominant Paradigm. More broadly, self-supervised learning has become a dominant paradigm in spoofing detection. Models such as Wav2Vec 2.0, XLS-R, HuBERT, and WavLM provide pretrained representations that often improve robustness when labeled spoofing data are limited [29]. An important finding in this literature is that intermediate SSL layers frequently capture low-level acoustic cues relevant to spoofing better than the final semantic layers, which motivated multi-layer aggregation strategies such as Sensitive Layer Selection [32].

Loss Design, Generalization, and Realistic Evaluation. Another important line of work treats anti-spoofing as a compact bona fide modeling problem rather than as a standard closed-set classification task. This motivates losses such as OC-Softmax and later extensions that encourage tighter bona fide representations and better generalization to unseen attacks [33], [34]. At the same time, the ASVspoof benchmarks have evolved toward increasingly realistic conditions, with ASVspoof 5 introducing more diverse recording conditions, environments, and acoustic degradations [31], [35]. These developments reinforce a central message of the literature: strong benchmark performance is not sufficient on its own, and robust anti-spoofing requires attention to generalization, realistic evaluation, and interpretability.

Generalization and Interpretability. Despite strong progress, cross-domain robustness remains a major open problem. Detectors often exploit dataset-specific shortcut cues, such as background conditions or recording artifacts, rather than synthesis artifacts themselves, which can lead to substantial performance drops on in-the-wild data [13]. This has also motivated growing interest in interpretability, especially for strong but complex models such as AASIST3, whose decisions may rely on non-obvious branch specializations or generator-specific artifact patterns [36]. This interpretability perspective is directly relevant to our project and is developed further in the explainability section below and in the dedicated analysis section later in the report.

C. Explainability for Audio and Multimodal Models

Explainability methods are increasingly important in deepfake audio detection because high predictive performance alone is not sufficient when model decisions must also be interpretable and trustworthy. In our setting, explainability serves two complementary goals: identifying which acoustic regions or features drive the detector decision, and assessing whether the detector relies on meaningful synthesis artifacts rather than on spurious dataset-specific shortcuts. We therefore only position the main families of methods here and defer the technical details of the approaches used in our project to the explainability section.

Feature Attribution Methods. A first family of methods explains predictions by assigning an importance score to input features or feature groups. Model-agnostic approaches such as LIME and SHAP are particularly useful when the detector operates on interpretable acoustic descriptors or structured spectro-temporal representations, since they reveal which features contribute most to a local decision. In audio deepfake detection, such methods can help determine whether the model relies on plausible cues such as pitch instability, spectral oversmoothness, or unnatural temporal regularity.

Gradient-Based Localization. A second family of approaches relies on gradients to localize discriminative regions for a given prediction. Methods such as saliency maps or Grad-CAM are especially relevant for convolutional detectors operating on spectrogram-like inputs, because they highlight the time–frequency regions that most influence the decision. These methods are useful for visually inspecting whether the detector attends to localized spoofing artifacts or to irrelevant background patterns.

Global Sensitivity and Robustness Analysis. Beyond local explanations, global attribution methods aim to characterize the detector behavior over an entire distribution of inputs. Variance-based or sensitivity-based approaches are useful in this context because they quantify how strongly the output depends on different feature groups and whether interactions between features play an important role. Such analyses complement local explanations by revealing broader detector biases and failure modes.

Overall, the explainability literature suggests that no single method is sufficient on its own: local and global analyses are most informative when used together. This motivates our own approach, where we combine several complementary explainability tools and study them in relation to generator type, detector behavior, and generalization.

III. Methodology & Results

A. Generation

As outlined in Section II, we now move from a high-level literature overview to the concrete generation models used in our experiments. In this section, we describe the selected VC, TTA, and TTS systems in greater detail, focusing on the mechanisms that matter for dataset construction and downstream detection.

1) Voice-Conversion: Voice conversion (VC) seeks to synthesize an utterance $\tilde{x}$ that preserves the linguistic content of a source speech signal x^s while transferring the speaker identity of a target reference x^t. Recent diffusion-based VC methods operate in Mel-spectrogram space. Let X_{mel} denote the target Mel-spectrogram and let Z be a data-driven prior constructed from disentangled attributes extracted from the source and target utterances. The generic forward process is written as a mean-reverting SDE

$$dX_t = \frac{1}{2}\beta_t(Z - X_t)\,dt + \sqrt{\beta_t}\,dW_t,$$

which differs from standard zero-mean diffusion by centering the terminal distribution on Z rather than on 0. The reverse dynamics are parameterized by a score network,

$$d\hat{X}_t = \left[\frac{1}{2}(Z - \hat{X}_t) - s_\theta(\hat{X}_t, Z, \mathbf{c}, t)\right]\beta_t\,dt + \sqrt{\beta_t}\,d\bar{W}_t,$$

where $\mathbf{c}$ denotes the conditioning variables. The practical design problem is therefore the construction of the prior Z and the choice of the conditioning variables. DiffVC introduced this formulation for one-shot many-to-many VC with an average-voice prior and a fast maximum-likelihood sampling scheme, as shown in Fig. 1 [10]. DDDM-VC and Diff-HierVC extend this framework by replacing the single prior by richer attribute-based priors and by refining the conditioning mechanism [1], [2].

DDDM-VC DDDM-VC is motivated by the observation that a single denoiser is poorly suited to control all speech attributes simultaneously. The method therefore decomposes VC into disentangled components and assigns a dedicated denoising pathway to each of them [1]. Let c, p, and s denote respectively the content, pitch, and target-style representations. In practice, c is extracted from a perturbed waveform with an intermediate XLS-R representation, p is obtained from normalized F_0 through a VQ-VAE, and s is extracted from the target Mel-spectrogram [1], [37].

The source–filter encoder maps these variables to two Mel-level priors,

$$Z_{\text{src}} = E_{\text{src}}(p, s), \qquad Z_{\text{ftr}} = E_{\text{ftr}}(c, s),$$

which are constrained by

$$L_{\text{rec}} = \|X_{\text{mel}} - (Z_{\text{src}} + Z_{\text{ftr}})\|_1.$$

Unlike DiffVC, the prior is therefore not an average voice, but a detailed decomposition of the target Mel-spectrogram into source- and filter-related components, as illustrated in Fig. 2. This reduces information loss in the prior and improves pronunciation and speaker adaptation.

Each branch is then modeled by a mean-reverting diffusion process,

$$dX_{n,t} = \frac{1}{2}\beta_t(Z_n - X_{n,t})\,dt + \sqrt{\beta_t}\,dW_t, \qquad n \in \{\text{src}, \text{ftr}\},$$

with two coupled score networks in the reverse dynamics. The coupling preserves attribute specialization while restoring the final speech jointly. The model is trained with a diffusion loss and the reconstruction term,

$$L = L_{\text{diff}} + \lambda_{\text{rec}} L_{\text{rec}}.$$

The second contribution is *prior mixup*. During training, the speaker embedding is randomly replaced by another style embedding s_r to construct converted priors

$$Z^r_{\text{src}} = E_{\text{src}}(p, s_r), \qquad Z^r_{\text{ftr}} = E_{\text{ftr}}(c, s_r).$$

The decoder is then trained to restore speech from these mismatched priors. This reduces the train–inference gap, since inference also operates from converted rather than reconstruction priors. Relative to DiffVC, DDDM-VC introduces three modifications: a richer source–filter prior, disentangled denoisers, and prior mixup for robust zero-shot conversion [1], [10].

Diff-HierVC Diff-HierVC starts from a different limitation of previous VC systems: inaccurate pitch modeling directly degrades pronunciation, intonation, and speaker similarity. The method therefore promotes pitch generation to a full diffusion stage instead of using only normalized or quantized F_0 as an auxiliary condition [2]. Speech is first decomposed into content c, pitch information, and target style s. As in DDDM-VC, content is extracted from a perturbed waveform with XLS-R, while style is obtained from the target Mel-spectrogram [2], [37].

The first stage, *DiffPitch*, generates target-style F_0. Let Z_p denote the pitch prior produced by the pitch encoder. It is regularized by

$$L_{\text{pitch}} = \|X_p - Z_p\|_1,$$

and used in the diffusion process

$$dX_{p,t} = \frac{1}{2}\beta_t(Z_p - X_{p,t})\,dt + \sqrt{\beta_t}\,dW_t.$$

The reverse process learns a score network s_{θ_p} that reconstructs the target-style pitch contour from noisy states. The central difference with respect to DiffVC is therefore that pitch is no longer a fixed conditioning signal, but a generated variable.

The second stage, *DiffVoice*, synthesizes the Mel-spectrogram from content, generated F_0, and target style; the

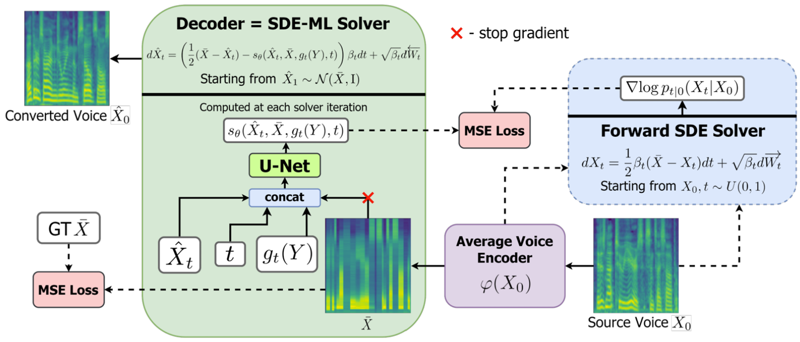


Figure 1. Architecture of DiffVC, a diffusion-based one-shot many-to-many voice conversion system. The model uses a data-driven prior and a mean-reverting diffusion process to synthesize a target Mel-spectrogram conditioned on source content and target speaker information.

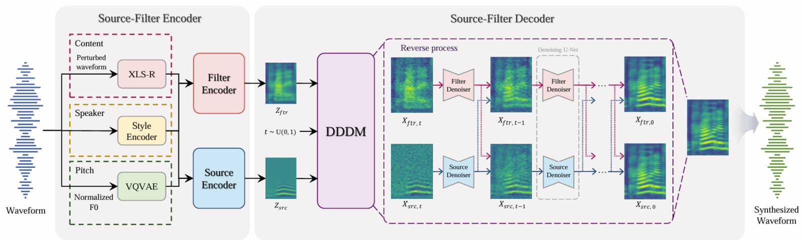


Figure 2. Architecture of DDDM-VC. The model decomposes voice conversion into disentangled source–filter priors and uses dedicated diffusion pathways for different speech attributes, together with prior mixup to reduce the train–inference mismatch in zero-shot conversion.

overall two-stage hierarchy is shown in Fig. 3. A source–filter encoder constructs the Mel prior

$$Z_m = Z_{\mathrm{src}} + Z_{\mathrm{ftr}}, \qquad Z_{\mathrm{src}} = E_{\mathrm{src}}(\hat{X}_p, s), \qquad Z_{\mathrm{ftr}} = E_{\mathrm{ftr}}(c, s),$$

with reconstruction constraint

$$L_{\mathrm{rec}} = \|X_{\mathrm{mel}} - Z_m\|_1.$$

DiffVoice then follows the same mean-reverting diffusion principle,

$$dX_{m,t} = \frac{1}{2}\beta_t(Z_m - X_{m,t})\,dt + \sqrt{\beta_t}\,dW_t,$$

with reverse denoising conditioned on Z_m and s.

Compared with DDDM-VC, the main contribution is hierarchical generation: first target-style pitch, then target-style Mel reconstruction from a source–filter prior. This improves pitch realism and zero-shot speaker adaptation while remaining within the same diffusion-based VC framework [1], [2].

2) Text-to-audio: Let x denote an audio waveform, X its Mel-spectrogram, and z_0 a compressed latent obtained by a VAE. In latent-diffusion TTA, one learns a conditional generative model

$$p_\theta(z_0 \mid c),$$

where c is a text-derived condition. The forward diffusion follows the standard Gaussian corruption process

$$q(z_t \mid z_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\, z_{t-1}, \beta_t I\right),$$
$$q(z_t \mid z_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\, z_0, (1-\bar{\alpha}_t) I\right).$$

and the reverse model is trained by noise prediction

$$\mathcal{L}_{\mathrm{diff}} = \mathbb{E}_{z_0,\epsilon,t}\left\|\epsilon - \epsilon_\theta\big(\sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon,\, t,\, c\big)\right\|_2^2.$$

The main design choice is therefore the definition of the condition c. AudioLDM uses a global CLAP embedding aligned across text and audio [3], [38]. AudioLDM 2 instead predicts a semantic sequence Y from the condition and uses this sequence to guide diffusion through cross-attention [4]. This second formulation is more expressive for temporally structured audio because Y is no longer a single global vector.

AudioLDM AudioLDM formulates TTA as latent diffusion conditioned on a shared text–audio embedding space [3]. Let $E_x = f_{\mathrm{audio}}(x)$ and $E_y = f_{\mathrm{text}}(y)$ be CLAP embeddings of the audio and text. Since CLAP aligns both modalities in the same latent space, the diffusion model is trained with audio embeddings only and sampled with text embeddings:

$$\mathcal{L}_{\mathrm{AudioLDM}} = \mathbb{E}_{z_0,\epsilon,t}\left\|\epsilon - \epsilon_\theta(z_t, t, E_x)\right\|_2^2,$$
$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon.$$

At inference, E_x is replaced by E_y. This removes the need to train the diffusion model on paired captions and audio at the generative stage.

The method contains four blocks, shown in Fig. 4. First, a pretrained CLAP model produces aligned text and audio embeddings [38]. Second, a convolutional VAE compresses the Mel-spectrogram into a latent tensor z_0, reducing the cost of diffusion [39]. Third, a UNet denoiser predicts the diffusion

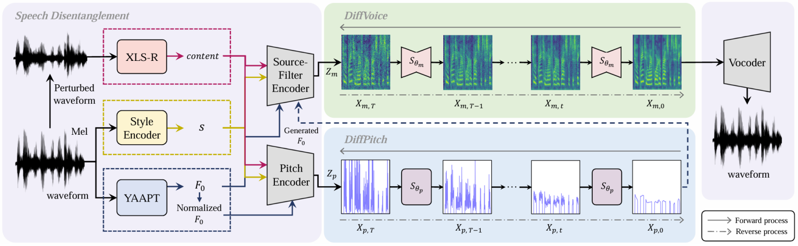


Figure 3. Architecture of Diff-HierVC. The model uses a hierarchical diffusion design: DiffPitch first generates a target-style pitch contour, and DiffVoice then synthesizes the Mel-spectrogram from content, generated pitch, and target speaker style.

noise in latent space. Fourth, a HiFi-GAN vocoder reconstructs the waveform from the decoded Mel-spectrogram [40].

Classifier-free guidance is used at sampling:

$$\hat{\epsilon}_\theta(z_t, t, E_y) = w\,\epsilon_\theta(z_t, t) + (1-w)\,\epsilon_\theta(z_t, t, E_y),$$

which improves text adherence. The key engineering point is that AudioLDM separates *semantic alignment* and *acoustic generation*: CLAP learns the former, latent diffusion learns the latter. The paper reports that this design outperforms DiffSound and AudioGen on AudioCaps, while also enabling zero-shot style transfer, inpainting, and super-resolution in the same latent space [3], [17], [18].

AudioLDM2 AudioLDM 2 replaces direct CLAP conditioning by a two-stage semantic–acoustic factorization, shown in Fig. 5 [4]. Let A denote an AudioMAE-based audio encoder, M a condition-to-semantics translator, and G a latent diffusion generator. The model introduces an intermediate sequence Y, called the *language of audio* (LOA), and writes generation as

$$H^0 = G \circ M : C \mapsto \hat{Y} \mapsto x,$$

while self-supervised training of the generator follows

$$H^1 = G \circ A : x \mapsto Y \mapsto \hat{x}.$$

Hence, G can be pretrained on audio only, whereas M is trained only to map the condition to LOA.

The LOA is extracted from AudioMAE features [41]. If $E \in \mathbb{R}^{T' \times F' \times D}$ is the AudioMAE output, a 2D pooling with factor λ yields a sequence

$$Y_\lambda \in \mathbb{R}^{L_\lambda \times D}, \qquad L_\lambda = \frac{T'F'}{\lambda^2}.$$

This sequence is then predicted autoregressively by GPT-2 from the condition C. For $Y = (y_1, \ldots, y_L)$, the translator optimizes

$$\max_\theta \; \Pr(y_1 \mid C; \theta) \prod_{l=2}^{L} \Pr(y_l \mid y_{<l}, C; \theta).$$

The condition C may combine CLAP text embeddings and FLAN-T5 embeddings in TTA, or other modality-specific encoders in other tasks.

The acoustic stage remains a VAE latent diffusion model, but conditioning is now a sequence rather than a global vector:

$$\mathcal{L}_{\mathrm{LDM}} = \mathbb{E}_{z_0,\epsilon,t} \left\| \epsilon - G\big(\sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon,\ t,\ Y\big) \right\|_2^2.$$

In contrast with AudioLDM, conditioning is injected through cross-attention over the LOA sequence. A probabilistic switcher is used during joint finetuning: the diffusion model is conditioned either on ground-truth LOA Y or GPT-predicted LOA $\hat{Y}$, with probabilities P_{gt} and P_{pred}. This reduces the train–inference mismatch between self-supervised generator pretraining and actual conditional sampling.

The practical difference with AudioLDM is therefore structural. AudioLDM directly maps a global semantic embedding to audio. AudioLDM 2 first predicts a structured semantic sequence, then synthesizes audio from it. This improves flexibility and extends the framework beyond TTA to text-to-music and Text-to-speech while keeping the same latent diffusion backbone [3], [4].

3) Text-to-speech: Let $y = (y_1, \ldots, y_L)$ denote an input text sequence, X_{mel} a target Mel-spectrogram, and s a speaker-conditioning signal or voice prompt. In our TTS branch, we consider two complementary model families: a flow-based parallel TTS model and a recent hierarchical zero-shot synthesizer. The first emphasizes fast and stable spectrogram generation from text, while the second emphasizes zero-shot speaker adaptation and expressive speech synthesis.

Glow-TTS. Glow-TTS formulates Text-to-speech as conditional generation with a normalizing flow and monotonic alignment search [19]. Let $h = E_{\mathrm{text}}(y)$ denote the encoded text representation. A flow-based decoder transforms the Mel-spectrogram into a latent variable

$$z = f_\theta(X_{\mathrm{mel}}; h),$$

and the conditional likelihood is written as

$$\log p_\theta(X_{\mathrm{mel}} \mid h) = \log p_Z(z) + \log \left| \det \frac{\partial f_\theta}{\partial X_{\mathrm{mel}}} \right|,$$

where p_Z is typically chosen as a standard Gaussian prior. The key difficulty is alignment between text tokens and acoustic frames. Glow-TTS resolves this with monotonic alignment search (MAS), which computes the most likely monotonic text–frame alignment directly during training rather than relying on an external autoregressive teacher. If $A^\star$ denotes the MAS alignment and d the induced token durations, the overall objective combines the flow likelihood with a duration-prediction loss,

$$\mathcal{L}_{\mathrm{GlowTTS}} = -\log p_\theta(X_{\mathrm{mel}} \mid h, A^\star) + \lambda_{\mathrm{dur}} \mathcal{L}_{\mathrm{dur}}(d, \hat{d}).$$

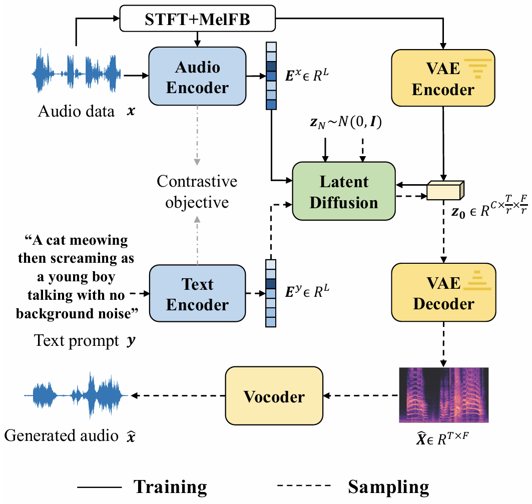


Figure 4. Architecture of AudioLDM. A pretrained CLAP model provides aligned text–audio embeddings, a VAE compresses Mel-spectrograms into a latent space, a latent diffusion model generates audio-conditioned latents, and a vocoder reconstructs the waveform.

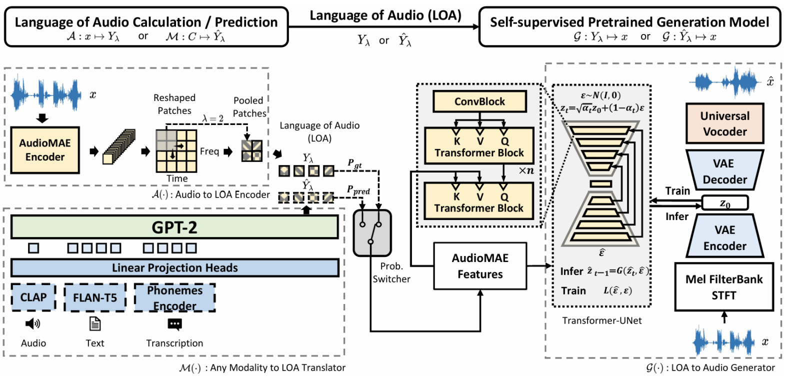


Figure 5. Architecture of AudioLDM 2. The model introduces a two-stage semantic–acoustic pipeline in which a condition-to-LOA translator predicts a semantic sequence and a latent diffusion generator synthesizes audio from that structured representation.

At inference time, the model first predicts durations from text, expands the text representation accordingly, samples a latent code z, and applies the inverse flow to synthesize a Mel-spectrogram, which is then converted to waveform by a vocoder. Relative to autoregressive TTS, the main advantages are parallel synthesis, stable monotonic alignment, and fast inference.

HierSpeech++. HierSpeech++ addresses a different regime: zero-shot and prompt-based speech synthesis, where the system must generate natural speech for unseen speakers from limited reference audio [11]. Its core idea is hierarchical generation from a semantic speech representation rather than directly from text alone. Let y denote the text input, p a prosody prompt, and s a voice prompt. A text-to-vec module predicts a self-supervised speech representation together with pitch-related information,

$$\hat{c} = G_{\text{vec}}(y, p), \qquad \hat{f}_0 = G_{F_0}(y, p),$$

where $\hat{c}$ is a semantic speech representation and $\hat{f}_0$ is the predicted pitch contour. A hierarchical decoder then generates speech conditioned on the semantic representation, pitch, and speaker prompt,

$$\hat{x}_{16\text{k}} = D(\hat{c}, \hat{f}_0, s).$$

HierSpeech++ further adds a speech super-resolution module that upsamples the generated waveform to higher fidelity,

$$\hat{x}_{48\text{k}} = U(\hat{x}_{16\text{k}}, s).$$

Conceptually, the model bridges semantic and acoustic levels through hierarchical variational inference: text and prompts first determine a higher-level speech representation, and waveform generation is deferred to a second stage. Compared with Glow-TTS, HierSpeech++ is less focused on classical parallel spectrogram generation and more focused on robust zero-shot speaker adaptation, expressive synthesis, and high-quality waveform generation from semantic, pitch, and speaker conditions. This makes it particularly relevant for deepfake generation, where speaker similarity and naturalness are central.

Although we implemented and reviewed TTS models as part of the broader project scope, the final experimental benchmark reported in this work focuses on VC and TTA only, since the TTS branch was not completed at sufficient scale for controlled generation, detection, and explainability analysis.

4) Dataset Construction: We constructed two complementary deepfake-audio detection datasets, based respectively on voice conversion and Text-to-audio synthesis. For the voice-conversion branch, we extracted a reproducible subset of real speech from the VCTK corpus and generated converted utterances with DDDM-VC and Diff-HierVC from deterministic source–target pairing manifests. For the Text-to-audio branch, we built a caption-conditioned dataset from Clotho and synthesized audio with AudioLDM and AudioLDM 2 using manifest-driven generation with fixed inference hyperparameters and seeded sampling. In both branches, the real/fake structure was propagated consistently through train/validation/test splits, generation outputs were tracked through structured manifests, and the final detection data were exported in both binary (real vs. fake) and generator-aware multiclass variants. Further implementation details for each generation pipeline are reported in Appendix C.

B. Generation Results

1) Voice-Conversion Results: We evaluate VC along two complementary dimensions: *content preservation* and *speaker transfer*. To keep the protocol fully automatic and aligned with the objective criteria emphasized in diffusion-based zero-shot VC, we report four metrics: CER, WER, EER, and SECS [1], [2]. CER and WER quantify how well the linguistic content of the source utterance is preserved after conversion, while EER and SECS quantify how closely the converted utterance matches the target speaker identity. We intentionally exclude subjective metrics such as nMOS and sMOS, since human listening studies were outside the scope of our student-scale and fully reproducible evaluation pipeline. We also do not report reconstruction-oriented measures such as MCD or F_0-based errors: although useful in some analyses, they are less directly aligned with our goal of comparing zero-shot VC models as deepfake generators. Finally, because our evaluation is fully automatic, CER and WER should be interpreted as reproducible *content-preservation proxies*, since transcript handling and ASR fallback can affect their absolute scale.

Table I

OVERALL VC RESULTS UNDER OUR EVALUATION SETUP. CER, WER, AND EER ARE REPORTED IN %; SECS IS REPORTED ON $[0, 1]$. LOWER IS BETTER FOR CER, WER, AND EER; HIGHER IS BETTER FOR SECS.

Model	CER↓	WER↓	EER↓	SECS↑
DDDM-VC	97.59	118.25	42.75	0.6009
Diff-HierVC	99.91	121.58	39.61	0.6034

On the aggregated results, the comparison is very close. DDDM-VC is marginally better on the content-related metrics, with lower CER and WER, whereas Diff-HierVC is marginally better on the speaker-similarity side, with lower EER and slightly higher SECS. Under our setup, this yields a small trade-off rather than a dominant winner: DDDM-VC slightly favors intelligibility and content consistency, while Diff-HierVC slightly favors speaker transfer.

This pattern is plausible in light of the inductive biases of the two methods. Both systems build on the DiffVC paradigm of mean-reverting diffusion for zero-shot VC [10], but they extend it in different directions. DDDM-VC introduces disentangled source–filter priors, dedicated denoising branches, and prior mixup, which are natural design choices for preserving linguistic structure under conversion [1]. Diff-HierVC instead introduces a hierarchical pipeline in which pitch is first generated and then injected into Mel-spectrogram generation, a design explicitly intended to improve prosody and target-speaker resemblance [2]. Our results are broadly consistent with this distinction, even though the observed gap is modest.

The split-wise results further suggest that the relative ordering is not entirely stable under our protocol. DDDM-VC is stronger on the train and test splits for CER/WER and remains competitive on SECS/EER, while Diff-HierVC is clearly stronger on the validation split. For completeness, the split-wise generation results are reported in Appendix D. In the main discussion, however, we treat the aggregated comparison in Table I as the primary result and use the split-wise analysis only as an indication of variability across subsets rather than as evidence for a definitive per-split winner.

These results should be interpreted strictly as a comparative evaluation under *our* setup, not as a reproduction of the original papers. Although the source works also compare DDDM-VC and Diff-HierVC on VCTK and generally suggest a slight advantage for Diff-HierVC in matched zero-shot settings, our protocol is not identical to theirs [1], [2]. Differences in pair construction, preprocessing, checkpoints, inference settings, transcript handling, and metric computation can affect both the absolute values and the ranking. The most defensible conclusion is therefore cautious: under our evaluation setting, DDDM-VC and Diff-HierVC behave as two strong and closely matched diffusion-based VC baselines, with a small trade-off between content preservation and speaker similarity, but without a clearly dominant winner.

2) Text-to-audio Results: For the Text-to-audio branch, we report three automatic objective metrics: *CLAP*, *FAD*, and *KL*. CLAP measures semantic alignment between the input caption and the generated audio in a joint text–audio embedding space, FAD measures the distributional distance between generated and real audio embeddings, and KL divergence measures the mismatch between the average tag distributions of generated and real clips [3], [4], [38]. Taken together, these metrics provide a compact view of prompt faithfulness, acoustic realism, and distributional consistency. As in the rest of this work, we restrict the evaluation to automatic criteria and do not report subjective measures such as OVL, REL, or MOS, which would require human listening studies and would be less compatible with our reproducible pipeline.

Table II
OVERALL TTA RESULTS UNDER OUR CORRECTED CLOTHO V2.1 EVALUATION SETUP. HIGHER IS BETTER FOR CLAP; LOWER IS BETTER FOR FAD AND KL.

Model	CLAP↑	FAD↓	KL↓	N
AudioLDM	0.322	3.062	0.606	500
AudioLDM 2	0.266	4.234	1.234	501

Under our evaluation protocol, AudioLDM outperforms AudioLDM 2 on all three reported metrics. It achieves a higher CLAP score (0.322 vs. 0.266), indicating stronger text–audio alignment in our setup, and lower FAD and KL values (3.062 vs. 4.234 for FAD; 0.606 vs. 1.234 for KL), indicating that its generated audio is also closer to the real-data distribution under the selected embedding and tagging backbones. On this basis, the comparative result within our framework is clear: under our Clotho v2.1-based evaluation, AudioLDM is the stronger of the two models on the retained objective metrics.

For completeness, split-wise results are reported in Appendix D-B. They are not the main basis for the conclusions in this subsection, but they do support the same overall pattern: AudioLDM remains ahead of AudioLDM 2 across train, validation, and test splits for CLAP, FAD, and KL. We therefore base the discussion primarily on the aggregated results in Table II, and use the split-wise breakdown only as supporting evidence for the stability of the observed ranking within our protocol.

This ranking differs from the trend emphasized in the source papers. In the original AudioLDM work, the main benchmark discussion is carried out on *AudioCaps*, with FD presented as the primary objective metric and FAD/KL reported as complementary measures [3]. In the AudioLDM 2 paper, again on *AudioCaps*, AudioLDM 2 variants are reported as stronger overall than earlier AudioLDM baselines, especially on distributional metrics such as FAD and KL [4]. Our comparison is therefore only partial: our values obtained in our pipeline should not be interpreted as benchmark-equivalent to the AudioCaps results reported in those papers.

A plausible explanation is that the observed ranking remains setup-dependent. Our experiments are conducted on *Clotho v2.1*, not AudioCaps, and therefore differ from the source papers in dataset distribution, caption style, preprocessing, checkpoints, generation settings, and metric-pipeline details. These differences are sufficient to affect both absolute values and relative ordering, so the present comparison should be read as a result under our own protocol rather than as a reproduction of the original benchmarks.

Rather than treating these objective generation metrics as a final validation on their own, we use them as a first comparative signal. A more practical cross-check will be provided in the detection section for both VC and TTA: under a naturalness-oriented criterion, the better generator should produce harder-to-detect deepfakes, and thus should induce lower downstream detector performance. We therefore return to this question later and interpret the generation-side ranking together with the detection results.

C. Detection

Section II positioned modern anti-spoofing methods at a high level. We now describe the three detector families used in our experiments: a lightweight MFCC-SVM baseline, spectrogram CNN baselines, and the stronger AASIST3 anti-spoofing architecture. Together, these models allow us to compare interpretability, in-domain performance, and robustness across different levels of detector complexity. The main question in this section is whether the differences observed on the generation side translate into differences in downstream detectability. In particular, we study whether VC and TTA induce distinct detection regimes, and whether these differences remain stable across lightweight baselines, spectrogram CNNs, and a stronger anti-spoofing architecture.

1) SVM Classifier: As a lightweight interpretable baseline, we train a Support Vector Machine (SVM) directly on handcrafted MFCC features extracted from each audio clip. The SVM seeks a linear decision boundary of maximum margin in a scaled feature space. Formally, let $\mathbf{x} \in \mathbb{R}^d$ be a feature vector and $y \in \{0, 1\}$ its binary label (0 = real, 1 = fake). The primal optimisation problem is

$$\min_{\mathbf{w},b,\boldsymbol{\xi}} \ \tfrac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \quad \text{s.t.} \quad y_i\big(\mathbf{w}^\top\mathbf{x}_i + b\big) \geq 1-\xi_i,\ \xi_i \geq 0,$$

where $\mathbf{w}$ and b define the separating hyperplane, ξ_i are slack variables that permit soft-margin violations, and $C > 0$ controls the trade-off between margin width and empirical loss. We use a linear kernel, $C = 1$, and set `class_weight=balanced` to handle potential label imbalance.

Feature extraction. Each audio clip is resampled to 16 kHz and truncated or zero-padded to a fixed duration of 4 s. We compute $N = 13$ MFCC coefficients and represent every clip by the concatenation of their per-frame mean and standard deviation, yielding a 26-dimensional vector:

$$\mathbf{x} = \big[\overline{\mathrm{mfcc}}_1, \ldots, \overline{\mathrm{mfcc}}_{13},\ \sigma_{\mathrm{mfcc}_1}, \ldots, \sigma_{\mathrm{mfcc}_{13}}\big]^\top \in \mathbb{R}^{26}.$$

Features are standardised with a z-score scaler fitted on the training split before being passed to the SVM.

Results on VCDetection10K. The dataset is split 80/10/10 into train, validation, and test sets (8 000 / 1 000 / 1 000 samples, balanced classes). The trained linear SVM uses only 201 support vectors in total (99 real, 102 fake), confirming that the two classes are almost linearly separable in MFCC space. Classification performance is summarised in Table III. The model reaches near-perfect accuracy on all three splits, with a ROC-AUC of 0.9998 on the test set. This result is consistent with AASIST3, and confirms that voice-conversion artefacts leave strong statistical traces in low-level cepstral statistics.

Results on TTA. The same pipeline is applied to the TTA dataset, loading real audio from the Clotho v2.1 subset and fake audio from AudioLDM1 and AudioLDM2 generations. Features are extracted and scaled identically. The SVM follows the same training procedure with an 80/10/10 split. Performance drops markedly compared to the VC setting: the

Table III
SVM PERFORMANCE ON VCDETECTION10K.

Split	Accuracy	Macro-F1	ROC-AUC
Train	0.993	0.993	0.9995
Validation	0.990	0.990	0.9993
Test	0.996	0.996	0.9998

Table IV
SVM PERFORMANCE ON THE TTA DATASET.

Split	Accuracy	Macro-F1	ROC-AUC
Train	0.756	0.753	0.8043
Validation	0.715	0.714	0.7975
Test	0.760	0.759	0.8107

model achieves 76.0 % accuracy and a ROC-AUC of 0.8107 on the test set (Table IV), using 986 support vectors — nearly five times more than in the VC case — reflecting a much less separable feature space. This confirms that Text-to-audio models produce ambient soundscapes whose cepstral statistics overlap substantially with natural recordings, whereas voice-conversion artefacts leave stronger and more consistent traces in low-level MFCC statistics. The performance gap mirrors the trend observed with AASIST3, where TTA consistently yielded weaker detection performance than VC.

2) Spectrogram CNN Baselines: As intermediate baselines between the MFCC-SVM and AASIST3, we consider spectrogram-based convolutional neural networks, focusing on VGG16 and ResNet18. This choice is motivated by two observations. First, spectrogram CNNs provide a natural extension from hand-crafted cepstral features to learned time–frequency representations. Second, both VGG-style and residual CNN backbones have long been strong image models and have also been used in audio-classification and anti-spoofing settings [21]–[23].

Let $x \in \mathbb{R}^T$ denote an input waveform. We first convert x into a time–frequency image

$$S = \Psi(x) \in \mathbb{R}^{F \times \tau},$$

where $\Psi(\cdot)$ denotes the spectrogram preprocessing pipeline. In practice, S is normalized and resized to a fixed spatial resolution before being passed to the network. This representation allows the detector to operate directly on local spectral textures, harmonic structure, and time–frequency artifacts that may distinguish bona fide from spoofed speech.

A spectrogram CNN detector can then be written as

$$h = E_{\text{cnn}}(S), \qquad \ell = C(h),$$

where E_{cnn} denotes the convolutional backbone and $C(\cdot)$ the final classification head. In the VGG16 case, E_{cnn} consists of stacked 3×3 convolutions and max-pooling operations that progressively transform the input spectrogram into higher-level feature maps [21]. In the ResNet18 case, feature extraction is performed through residual blocks, which facilitate deeper optimization through skip connections. After the final convolutional stage, the feature tensor is aggregated and mapped to class logits.

In the binary setting, the model predicts whether an input is bona fide or fake:

$$\ell \in \mathbb{R}^2, \qquad \hat{y} = \arg\max_k \ell_k.$$

Training is performed with the standard cross-entropy objective,

$$\mathcal{L}_{\text{CE}} = -\sum_{k=1}^{2} \mathbf{1}[y = k] \log \frac{\exp(\ell_k)}{\sum_j \exp(\ell_j)}.$$

In practice, both CNN baselines are fine-tuned in two stages: first, only the classification head is optimized while most pretrained convolutional parameters remain frozen; second, the last high-level convolutional block is unfrozen and jointly fine-tuned with a smaller learning rate. This strategy aims to preserve generic visual priors while adapting the upper layers to spectrogram-specific deepfake artifacts.

Compared with AASIST3, these spectrogram CNN baselines have a simpler inductive bias. They do not rely on a pretrained self-supervised speech encoder or on explicit graph-based spectro-temporal reasoning, but instead learn directly from spectrogram images through standard 2D convolutions. They therefore serve as useful intermediate baselines: more expressive than the MFCC-SVM, but still substantially simpler than modern anti-spoofing architectures such as AASIST3.

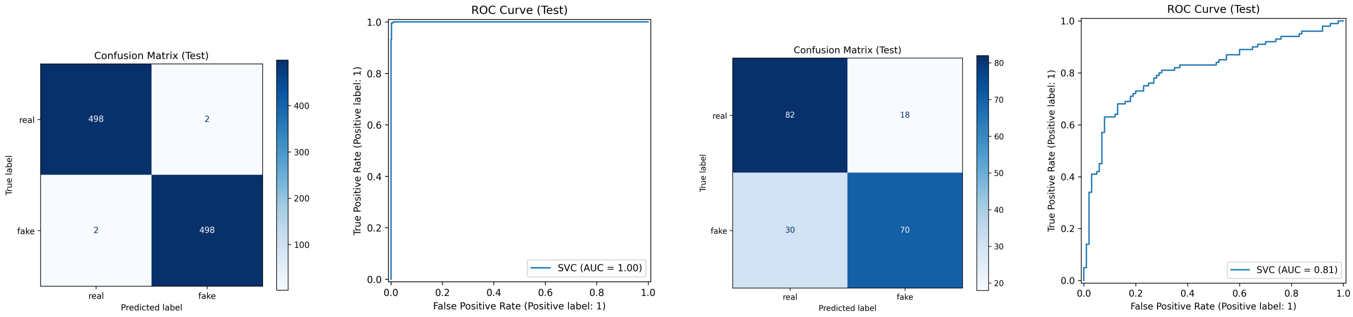
3) AASIST: **Audio Deepfake Detection.** Let $x \in \mathbb{R}^T$ denote an input waveform and let f_θ be an audio deepfake detector that maps speech to class logits,

$$\ell = f_\theta(x), \qquad \hat{y} = \arg\max_k \ell_k.$$

In our system, f_θ is instantiated by AASIST3, a detector that combines a pretrained self-supervised front-end with convolutional and graph-attention back-end modules. Rather than relying on a single global representation, the model explicitly decomposes speech features into spectral and temporal graph views and aggregates them through multiple inference branches before classification. The overall design is motivated by the observation that spoofing traces are often localized and non-stationary across time-frequency structure, so detection benefits from structured feature aggregation rather than from a single global pooling operator.

The practical problem is not to redesign the detector backbone itself, but to adapt it to our project-specific dataset and evaluation protocol. We therefore build the detection module around two settings: binary real-versus-fake detection and generator-aware multiclass detection. The common backbone is trained and evaluated on VCDetection10K, where each sample is normalized to mono 16 kHz speech and segmented to a fixed input length. The dataset supports both binary labels and generator-aware labels, enabling the detector to operate either as a standard anti-spoofing classifier or as a generator-discriminative classifier.

AASIST3 Backbone. The AASIST3 detector used in this project follows a self-supervised front-end plus graph-based aggregation design. Let $x \in \mathbb{R}^T$ denote the input waveform.



(a) CM — VCDetection10K (b) ROC — VCDetection10K (c) CM — TTA (d) ROC — TTA

Figure 6. SVM test-set confusion matrices (CM) and ROC curves for VCDetection10K (a, b) and the TTA dataset (c, d).

The detector first extracts SSL features with a pretrained Wav2Vec2 encoder,

$$H_{\text{ssl}} = E_{\text{w2v}}(x),$$

where $H_{\text{ssl}} \in \mathbb{R}^{L\times d_{\text{ssl}}}$ is a sequence of contextual speech representations. Since these SSL features are high-dimensional, they are projected through a bridge module before downstream processing:

$$\tilde{H} = B(H_{\text{ssl}}).$$

The projected representation is then reshaped and processed by a residual convolutional encoder,

$$E = E_{\text{enc}}(\tilde{H}),$$

where E denotes the intermediate spectro-temporal feature tensor.

AASIST3 then constructs two complementary graph views from E. The first is a spectral graph obtained by aggregating along the temporal dimension,

$$G_s = \text{Pool}_t(E),$$

and the second is a temporal graph obtained by aggregating along the frequency dimension,

$$G_t = \text{Pool}_f(E).$$

These two graph representations are further refined by graph-attention modules,

$$Z_s = \text{GAT}_s(G_s), \qquad Z_t = \text{GAT}_t(G_t),$$

so that the model can capture spoofing cues distributed across different frequency bands and time regions. This spectro-temporal decomposition is important because deepfake artifacts are often sparse and non-stationary, rather than uniformly distributed over the entire utterance.

After graph reasoning, AASIST3 applies several parallel inference branches to model higher-order interactions between the spectral and temporal streams. Their outputs are aggregated together with a global master representation,

$$h = \text{Agg}(Z_s, Z_t, m),$$

where m denotes the branch-wise global token and h is the final hidden representation used for classification. The detector output is then computed as

$$\ell = C(h),$$

where $C(\cdot)$ denotes the final classifier head. In the binary setting, $\ell \in \mathbb{R}^2$, while in the generator-aware multiclass setting, $\ell \in \mathbb{R}^3$.

This backbone is particularly suitable for deepfake audio detection because modern generative systems often preserve high-level linguistic content while introducing localized vocoder artifacts, phase inconsistencies, or generator-specific spectral regularities. The detector therefore relies on learned structural aggregation rather than handcrafted spoofing features, which makes it adaptable to different fake-audio families within a unified framework.

The architecture in Fig. 7 summarizes the full processing pipeline from waveform input to final decision. It highlights the staged transformation from SSL representations to graph-structured spectro-temporal reasoning, which is the key distinction between AASIST-style detectors and simpler convolution-only baselines.

Figure 8 further emphasizes the local computation pattern inside the KAN-enhanced graph-attention module. In practice, this component provides a more flexible nonlinear mapping than standard linear projections and contributes to the stronger representation capacity of AASIST3 under challenging spoofing conditions.

Binary Detection. The first detection task is binary anti-spoofing, in which the detector distinguishes bona fide speech from fake speech. Let $\ell = f_\theta(x) \in \mathbb{R}^2$ be the binary logits and let $y \in \{0, 1\}$ be the aligned target. Training uses the cross-entropy objective

$$\mathcal{L}_{\text{bin}} = -\sum_{k=1}^{2} \mathbf{1}[y = k] \log \frac{\exp(\ell_k)}{\sum_j \exp(\ell_j)}.$$

This binary setting provides the primary detection baseline and measures whether the pretrained anti-spoofing backbone can be transferred successfully to our generated-audio data.

The model is initialized from pretrained AASIST3 weights rather than trained from scratch. This choice reflects the fact

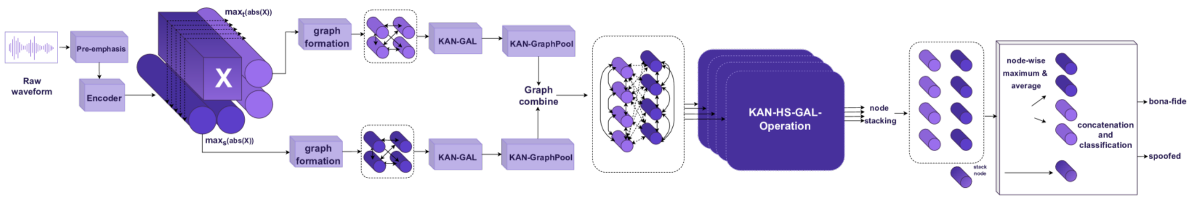


Figure 7. Overall architecture of AASIST3. The model combines an SSL front-end, bridge projection, residual convolutional encoding, spectro-temporal graph construction, and multi-branch graph-attention inference before final classification.

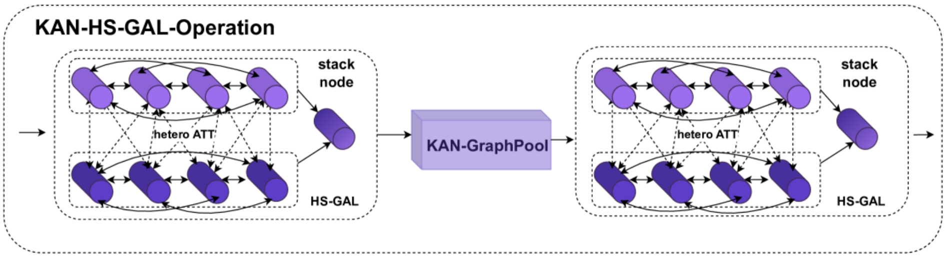


Figure 8. KAN-HS-GAL module in AASIST3. The module integrates Kolmogorov-Arnold nonlinear transformations with heterogeneous spectro-temporal graph attention for artifact-sensitive feature modeling.

that anti-spoofing is already a mature discriminative problem and that pretrained detectors encode useful priors about speech authenticity. Fine-tuning therefore focuses on adapting this prior to the distribution induced by our own generation pipelines.

Two-Stage Fine-Tuning. Binary fine-tuning is performed in two stages. In the first stage, the SSL front-end is frozen and only the downstream detector parameters are optimized; in the second stage, the full model is unfrozen and jointly optimized with a smaller learning rate. Writing the model parameters as $(\theta_{\mathrm{ssl}}, \theta_{\mathrm{det}})$, the training schedule is

$$\text{Stage 1:} \min_{\theta_{\mathrm{det}}} \mathcal{L}_{\mathrm{bin}}, \qquad \text{Stage 2:} \min_{\theta_{\mathrm{ssl}}, \theta_{\mathrm{det}}} \mathcal{L}_{\mathrm{bin}}.$$

This staged design reduces instability at the beginning of training while still allowing final task-specific adaptation of the self-supervised representation.

The practical role of this strategy is similar to the staged constructions used in the generation section: a strong pretrained component is preserved initially, and task-specific refinement is introduced only after the downstream layers have adapted to the new data distribution. In the present case, however, the objective is discriminative rather than generative.

Binary Evaluation. Binary detection is evaluated with standard anti-spoofing metrics derived from scalar detection scores. Let $s(x)$ denote the spoofing score extracted from the detector output. We report Equal Error Rate (EER), minimum Detection Cost Function (minDCF), and CLLR. These metrics are threshold-sensitive and therefore more informative than plain accuracy for authentication-style tasks. They characterize not only whether the detector separates real from fake speech, but also whether its score distribution supports reliable operating points under different decision thresholds.

Generator-Aware Multiclass Detection. The second task extends binary anti-spoofing to generator-aware multiclass classification. Rather than only deciding whether an utterance is fake, the detector is trained to identify which voice-conversion system produced it. Let

$$y \in \{0, 1, 2\}$$

denote the labels for real speech, DDDM-VC fake speech, and Diff-HierVC fake speech, and let

$$\ell = f_\theta(x) \in \mathbb{R}^3$$

denote the corresponding logits. The multiclass objective is again cross-entropy,

$$\mathcal{L}_{\mathrm{multi}} = -\sum_{k=1}^{3} \mathbf{1}[y = k] \log \frac{\exp(\ell_k)}{\sum_j \exp(\ell_j)}.$$

This problem is strictly harder than binary detection, since the detector must capture not only authenticity cues but also generator-specific artifact structure.

Generator-aware detection is particularly relevant in our setting because the fake samples are produced by two different VC pipelines described in Section III-A. A successful multiclass detector therefore indicates that the learned representation contains information beyond a simple real/fake boundary and retains structured traces of the synthesis process itself.

Multiclass Head Adaptation. The multiclass model is obtained by adapting the pretrained AASIST3 detector to a new output dimension. Since the original checkpoint is binary, only the compatible parameters are transferred, while incompatible layers, in particular the final classifier, are reinitialized. If Θ_{pre} denotes the pretrained parameter set and Θ_{new} the multiclass model parameters, the initialization can be viewed as

$$\Theta_{\mathrm{new}}^{(0)} = \{\theta \in \Theta_{\mathrm{pre}} : \mathrm{shape}(\theta) \text{ matches}\} \cup \{\theta_{\mathrm{new}} : \text{otherwise}\}.$$

This partial transfer preserves the pretrained representation while adapting the decision layer to the new label space.

Conceptually, this is analogous to transferring a feature extractor across related tasks while relearning only the task-specific boundary. The multiclass detector thus reuses the anti-spoofing backbone as a general speech-authenticity encoder and specializes its final layer for generator discrimination.

Multiclass Evaluation. Multiclass detection is evaluated with accuracy, macro-F1, and the confusion matrix. Let $C \in \mathbb{N}^{3\times 3}$ be the confusion matrix with entries C_{ij} counting samples of true class i predicted as class j. Accuracy is

$$\text{Acc} = \frac{\sum_i C_{ii}}{\sum_{i,j} C_{ij}},$$

and macro-F1 averages the class-wise F1 scores. These metrics are complemented by per-class precision and recall, which reveal whether the detector separates the two fake generators symmetrically or whether one generator is consistently more confusable than the other. The multiclass setting therefore provides a finer diagnostic of representation quality than the binary task alone.

In summary, the detection module uses AASIST3 as a common backbone and studies two complementary settings: binary anti-spoofing and generator-aware multiclass detection. The binary task measures the core ability to distinguish real from fake speech, while the multiclass task asks whether generator identity can also be recovered from the same detection framework.

Together, these components form the detection counterpart to the generation pipelines described earlier. The generation module produces controlled fake-audio families with traceable provenance, while the detection module evaluates whether a modern anti-spoofing architecture can detect them and separate their generator signatures within a unified discriminative framework.

4) Per-Dataset Evaluation: To complement the methodological description above, we report per-dataset validation and test performance for both binary and multiclass detection. We additionally visualize the optimization dynamics through training-loss curves, which show the progressive decrease of the objective during fine-tuning on each dataset and task setting. Figure 9 summarizes the loss trajectories for VCDetection10K and TTA under binary and multiclass training.

The results reveal a clear dataset-dependent gap. On VCDetection10K, binary detection is nearly perfect on validation and remains extremely strong on test, with EER close to zero and multiclass accuracy above 99%. The corresponding confusion matrices also indicate that the detector separates real speech and the two VC generators almost perfectly. On the TTA dataset, by contrast, the binary EER remains around 16.4% on both validation and test, and multiclass accuracy is about 71%, with the third class showing substantially lower recall than the first two. This suggests that the artifacts introduced by Text-to-audio generation are less easily separable than those produced by the VC pipelines considered here.

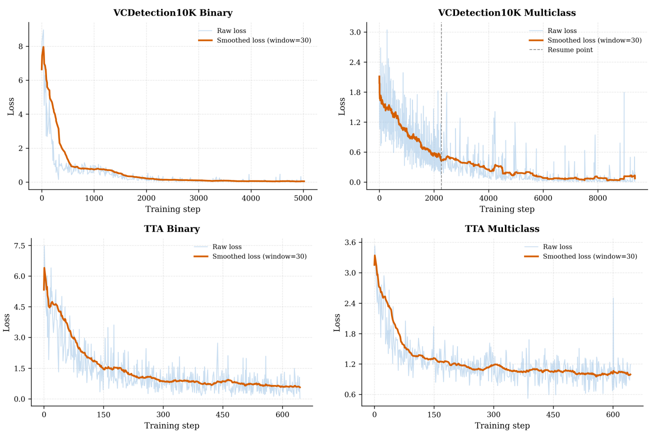


Figure 9. Training loss curves for AASIST fine-tuning. Top row: VCDetection10K binary and multiclass. Bottom row: TTA binary and multiclass. In all cases the smoothed loss decreases steadily, indicating stable optimization; however, the final downstream performance remains strongly dataset-dependent, with VCDetection10K substantially easier than TTA.

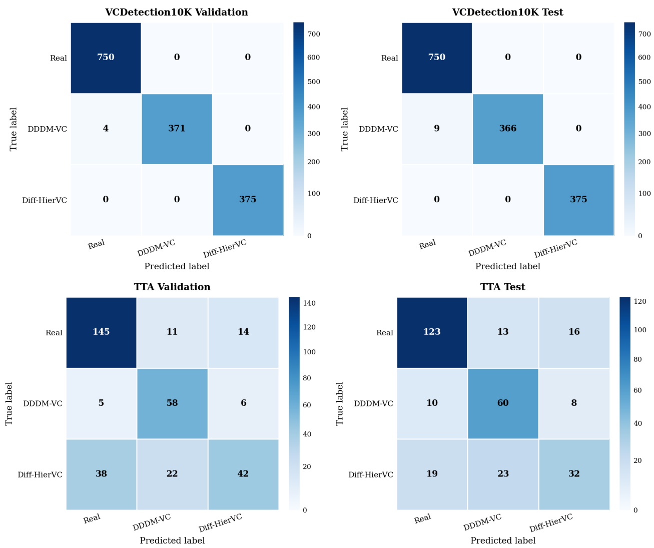


Figure 10. Confusion-matrix heatmaps for multiclass AASIST evaluation. Top row: VCDetection10K validation and test. Bottom row: TTA validation and test. The VCDetection10K matrices are strongly diagonal, while the TTA matrices show more pronounced cross-class confusion, especially for the third class.

For completeness, the confusion matrices are reported below:

These results indicate that AASIST transfers very effectively to the VC-based detection setting, where generator artifacts appear highly structured and discriminative, but is noticeably less effective on TTA-generated audio, where the boundary between real and fake audio is less sharply defined and class-specific signatures are weaker.

5) CNN Spectrogram Baselines and Comparison with AASIST3: To complement AASIST3, we evaluate two transfer-learning CNN baselines on Mel-spectrogram images: VGG16 and ResNet18. In both implementations, audio clips are converted to fixed-length Mel-spectrograms, resized to a common

Table V
VALIDATION PERFORMANCE OF AASIST ON VCDETECTION10K AND TTA UNDER BINARY AND MULTICLASS SETTINGS.

Dataset	Task	EER / Accuracy	minDCF	CLLR / Macro-F1	Additional Results
VCDetection10K	Binary	EER = 0.0000	0.0000	0.1995	–
VCDetection10K	Multiclass	Accuracy = 0.9973	–	Macro-F1 = 0.9973	Recall = [1.0000, 0.9893, 1.0000]
TTA	Binary	EER = 0.1642	0.3957	1.0717	–
TTA	Multiclass	Accuracy = 0.7185	–	Macro-F1 = 0.6824	Recall = [0.8529, 0.8406, 0.4118]

Table VI
TEST PERFORMANCE OF AASIST ON VCDETECTION10K AND TTA UNDER BINARY AND MULTICLASS SETTINGS.

Dataset	Task	EER / Accuracy	minDCF	CLLR / Macro-F1	Additional Results
VCDetection10K	Binary	EER = 0.0013	0.0025	0.2054	–
VCDetection10K	Multiclass	Accuracy = 0.9940	–	Macro-F1 = 0.9940	Recall = [1.0000, 0.9760, 1.0000]
TTA	Binary	EER = 0.1645	0.4572	1.1278	–
TTA	Multiclass	Accuracy = 0.7072	–	Macro-F1 = 0.6637	Recall = [0.8092, 0.7692, 0.4324]

image size, and used for binary classification (real vs. fake). For both TTA and VC data, generator-specific fake subsets (fake1 and fake2) are mapped to the same fake label, so the task remains binary.

Both CNN pipelines use two-stage fine-tuning. In Phase 1, most pretrained ImageNet parameters are frozen and only the classification head is optimized. In Phase 2, the last high-level block is unfrozen and jointly fine-tuned with a smaller learning rate. This strategy aims to preserve generic visual priors while adapting upper layers to spectrogram artifacts.

Evaluation protocol. For a direct and compact comparison, we report three representative binary settings available for both backbones: (i) in-domain TTA (train/test on TTA mixed), (ii) in-domain VC (train/test on VC mixed), and (iii) cross-domain transfer (train on VC mixed, test on TTA mixed). Table VII summarizes test performance.

CNN baseline observations. Both backbones are strong and nearly identical on in-domain TTA. By contrast, in-domain VC is substantially harder: both models keep very high real recall but much lower fake recall, indicating a bias toward bona fide predictions for a significant fraction of VC fakes. Cross-domain transfer from VC to TTA degrades sharply, with performance near chance level, confirming poor domain robustness for these spectrogram-only CNN baselines.

Important qualitative plots. Figures 11 and 12 show the corresponding confusion matrices: near-diagonal behavior for in-domain TTA, stronger fake misses for in-domain VC, and large confusion under VC→TTA transfer.

Comparison against AASIST3. AASIST3 was evaluated with anti-spoofing metrics in binary mode (EER/minDCF/CLLR) and with Accuracy/Macro-F1 in multiclass mode. Since metric families differ from the CNN binary reports, we compare trends rather than absolute score equivalence. Table VIII summarizes the main test-set outcomes.

Interpretation. The comparison reveals complementary behavior. AASIST3 is extremely strong on VCDetection10K, while the spectrogram CNN baselines are substantially weaker on VC. Conversely, both VGG16 and ResNet18 perform strongly on in-domain TTA binary detection under our protocol. Across architectures, robustness remains the core limi-

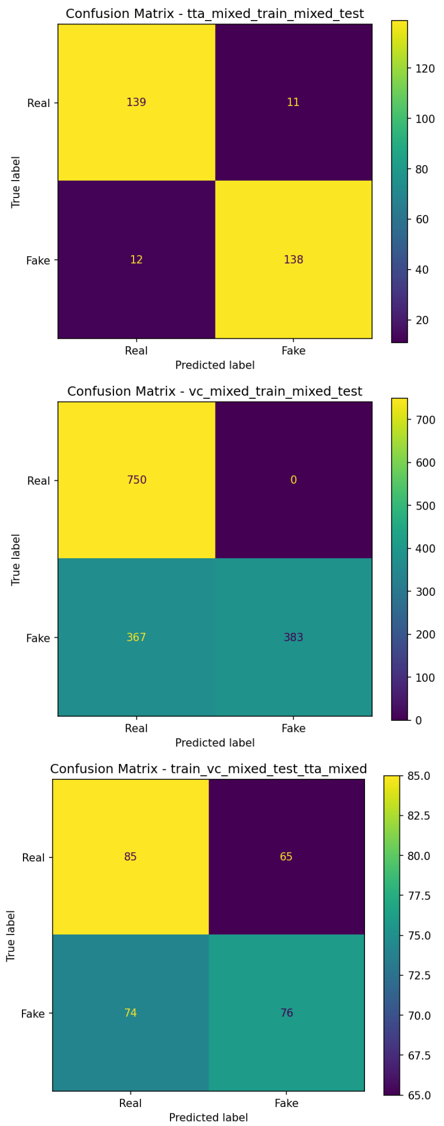


Figure 11. VGG16 confusion matrices (left to right): in-domain TTA, in-domain VC, and cross-domain VC→TTA.

Table VII
BINARY DEEPFAKE DETECTION RESULTS FOR SPECTROGRAM CNN BASELINES. HIGHER IS BETTER FOR ACCURACY, MACRO-F1, AND RECALLS; LOWER IS BETTER FOR LOSS.

Experiment	Backbone	Accuracy	Macro-F1	Loss	Real Recall	Fake Recall
TTA mixed → TTA mixed	VGG16	0.9233	0.9233	0.2218	0.9267	0.9200
TTA mixed → TTA mixed	ResNet18	0.9233	0.9233	0.1897	0.9067	0.9400
VC mixed → VC mixed	VGG16	0.7553	0.7398	1.1284	1.0000	0.5107
VC mixed → VC mixed	ResNet18	0.7400	0.7233	0.8876	0.9853	0.4947
VC mixed → TTA mixed	VGG16	0.5367	0.5362	2.4636	0.5667	0.5067
VC mixed → TTA mixed	ResNet18	0.4533	0.4492	1.2075	0.5400	0.3667

Table VIII
SUMMARY COMPARISON BETWEEN CNN BASELINES AND AASIST3 ON TEST SETS. FOR AASIST3, LOWER IS BETTER FOR EER/MINDCF/CLLR.

Dataset/Setting	Model	Metric Set	Values	MC Acc.	MC F1	Trend
VCDetection10K	AASIST3	EER/minDCF/CLLR	0.0013/0.0025/0.2054	0.9940	0.9940	Near-perfect VC detection
VC mixed (bin.)	VGG16	Acc./Macro-F1	0.7553/0.7398	–	–	Many fake misses
VC mixed (bin.)	ResNet18	Acc./Macro-F1	0.7400/0.7233	–	–	Many fake misses
TTA (bin.)	AASIST3	EER/minDCF/CLLR	0.1645/0.4572/1.1278	–	–	Harder than VC for AASIST3
TTA mixed (bin.)	VGG16	Acc./Macro-F1	0.9233/0.9233	–	–	Strong in-domain TTA detection
TTA mixed (bin.)	ResNet18	Acc./Macro-F1	0.9233/0.9233	–	–	Strong in-domain TTA detection
TTA (multi.)	AASIST3	Acc./Macro-F1	–	0.7072	0.6637	Moderate generator separation
Cross-domain VC→TTA (bin.)	VGG16	Acc./Macro-F1	0.5367/0.5362	–	–	Poor transfer
Cross-domain VC→TTA (bin.)	ResNet18	Acc./Macro-F1	0.4533/0.4492	–	–	Poor transfer

tation: cross-domain transfer is weak for CNNs, and AASIST3 also shows a marked performance drop from VC to TTA. Overall, these results support the same high-level conclusion as the rest of the detection analysis: current detectors can achieve excellent in-domain performance, but domain and generator shifts remain a central bottleneck.

The detection results raise a natural follow-up question: which cues do the models actually rely on, and are these cues consistent with the observed differences between VC and TTA? The purpose of the explainability analysis is therefore not only to interpret individual predictions, but also to understand the source of the performance gap across generator families.

D. Explainability

After the brief overview given in Section II, we now detail the explainability methods considered in this project and how they are applied to deepfake-audio detection.

1) SHAP-Based Explanations: SHAP provides a local additive decomposition of the detector output and is particularly useful when the deepfake classifier operates on structured acoustic descriptors or on aggregated spectro-temporal features rather than on raw waveform samples. Let $f(x)$ denote the spoofing logit or posterior score for an input representation $x \in \mathbb{R}^d$. SHAP approximates the prediction around x by

$$f(x) \approx \phi_0 + \sum_{i=1}^{d} \phi_i,$$

where ϕ_i is the Shapley attribution associated with feature i and $\phi_0 = \mathbb{E}[f(X)]$ is the baseline output [7]. Each coefficient is uniquely determined by the Shapley axioms (efficiency, symmetry, dummy, linearity) and admits the combinatorial form

$$\phi_i = \sum_{S \subseteq \{1,\dots,d\}\setminus\{i\}} \frac{|S|!\,(d-|S|-1)!}{d!} \left[f_S(S \cup \{i\}) - f_S(S)\right],$$

where $f_S(S) = \mathbb{E}[f(X) \mid X_S = x_S]$ marginalizes over the complement of S.

In deepfake audio detection, this decomposition is most informative when x is defined at an interpretable level (e.g., LFCC coefficients). Under this formulation, SHAP reveals whether the detector relies on physically plausible cues such as pitch micro-instability, formant inconsistency, or unnatural temporal smoothness, rather than on dataset-specific confounders. Aggregating local attributions across the evaluation corpus provides a global perspective: the mean absolute Shapley value $\bar{\Phi}_i = \frac{1}{N}\sum_n |\phi_i^{(n)}|$ ranks features by average importance, while conditioning on deepfake subsets isolates generator-specific signatures. Comparing these distributions across generation methods, diffusion-based voice conversion versus autoregressive TTS for instance, directly informs robustness and generalization diagnostics.

2) LIME-Based Local Interpretability: LIME provides a model-agnostic local explanation by fitting a simple surrogate model in the neighborhood of a given input [8]. Let f denote the deepfake detector and $x \in \mathbb{R}^d$ an input representation. LIME defines an interpretable domain of binary masks $z \in \{0,1\}^{d'}$, where each coordinate indicates the presence or absence of a feature group. A perturbed input x_z is obtained by masking the corresponding components, and the surrogate is trained by solving

$$\xi(x) = \arg\min_{g \in \mathcal{G}} \sum_{z \sim \pi_x} \pi_x(z) \left[f(x_z) - g(z)\right]^2 + \Omega(g),$$

where $\mathcal{G}$ is a class of interpretable models (typically sparse linear functions), $\pi_x(z) = \exp\bigl(-D(x, x_z)^2/\sigma^2\bigr)$ is a proxim-

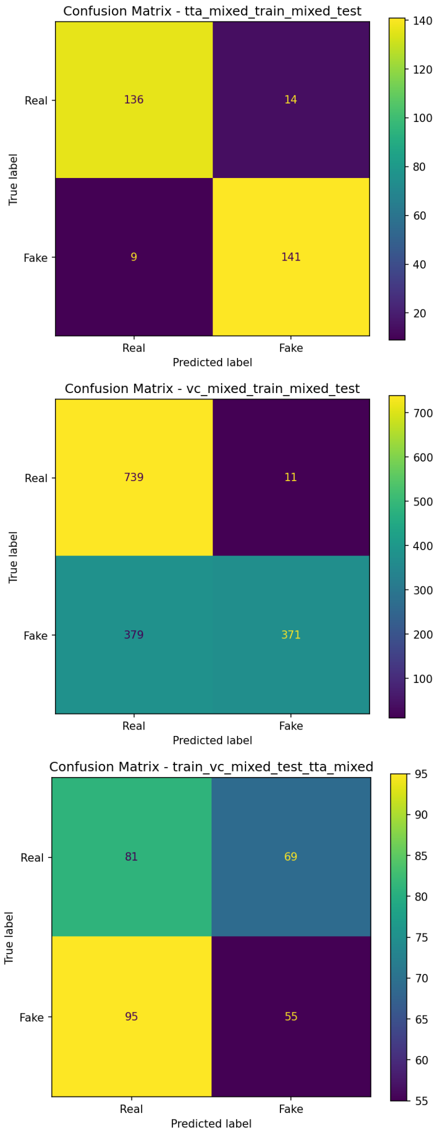


Figure 12. ResNet18 confusion matrices (left to right): in-domain TTA, in-domain VC, and cross-domain VC→TTA.

ity kernel, and $\Omega(g)$ is a sparsity-inducing penalty [8]. The coefficients of the fitted surrogate then serve as local feature-importance scores.

In the audio deepfake setting, the main design choice is the definition of the interpretable domain. When the detector operates on spectrograms, z controls contiguous time-frequency patches; when it operates on hand-crafted descriptors, each feature naturally maps to one coordinate. This flexibility makes LIME applicable to heterogeneous detection pipelines without requiring access to model gradients, unlike Grad-CAM or saliency methods. The main practical limitation is sensitivity to the segmentation granularity and the masking strategy, both of which influence the quality of the local surrogate fit. Despite this, LIME remains a useful complement to SHAP when Shapley estimation is computationally prohibitive or when the detector architecture does not permit gradient-based analysis.

3) Grad-CAM and Saliency-Based Analysis: For convolutional or convolutional-attention detectors ingesting log-Mel spectrograms, Grad-CAM and saliency methods provide a direct localization of discriminative regions in the time–frequency plane. Given a target class score y^c and a feature tensor $A^k \in \mathbb{R}^{T' \times F'}$ at the last convolutional block, Grad-CAM computes channel weights

$$\alpha_k^c = \frac{1}{T'F'} \sum_{i=1}^{T'} \sum_{j=1}^{F'} \frac{\partial y^c}{\partial A_{ij}^k}$$

and produces the coarse localization map

$$L_{\text{Grad-CAM}}^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c \, A^k \right),$$

which can then be upsampled to spectrogram resolution [20]. In the audio deepfake setting, this map highlights which temporal segments and frequency bands drive the spoofing decision.

4) Variance-Based Attribution Methods: Variance-based attribution methods rest on a global sensitivity perspective in which importance is defined relative to an input distribution rather than to an infinitesimal perturbation around a single point. Let $f : \mathbb{R}^d \to \mathbb{R}$ denote the detector score and let $X = (X_1, \ldots, X_d) \sim p_X$ be a random input representation. Writing $Y = f(X)$, the law of total variance,

$$\mathrm{Var}(Y) = \mathrm{Var}(\mathbb{E}[Y \mid X_i]) + \mathbb{E}[\mathrm{Var}(Y \mid X_i)],$$

separates the output variance explained by X_i from the residual variability. Under the assumption that the coordinates of X are independent and $f \in L^2(p_X)$, the Hoeffding-Sobol functional ANOVA decomposition yields an orthogonal expansion

$$f(X) = f_0 + \sum_i f_i(X_i) + \sum_{i<j} f_{ij}(X_i, X_j) + \cdots,$$

where each component has zero mean and is orthogonal to all lower-order terms [9]. The Sobol first-order sensitivity index is then

$$S_i = \frac{\mathrm{Var}(\mathbb{E}[Y \mid X_i])}{\mathrm{Var}(Y)},$$

while the total-order index, capturing all effects involving X_i including interactions, is

$$S_i^T = \frac{\mathbb{E}[\mathrm{Var}(Y \mid X_{\sim i})]}{\mathrm{Var}(Y)},$$

where $X_{\sim i}$ denotes all variables except X_i. The gap $S_i^T - S_i$ directly measures the contribution of interactions involving feature i.

In the context of deepfake audio detection, variance-based attribution complements local methods in two respects. First, the indices are inherently global: they characterize the detector over an entire population of utterances rather than for a single input, making them suitable for auditing whether a classifier systematically relies on artifact-bearing frequency ranges. Second, the decomposition into first-order and interaction terms provides a principled diagnostic for model complexity. If total-order indices substantially exceed first-order indices for spectral and temporal features simultaneously, the detector captures cross-domain interactions that no single-feature explanation can represent.

More broadly, recent work has emphasized that audio explainability remains less mature than vision explainability, and that deepfake-audio detection in particular still faces an explainability gap, especially for strong transformer-based detectors and real-world generalization settings [14], [15].

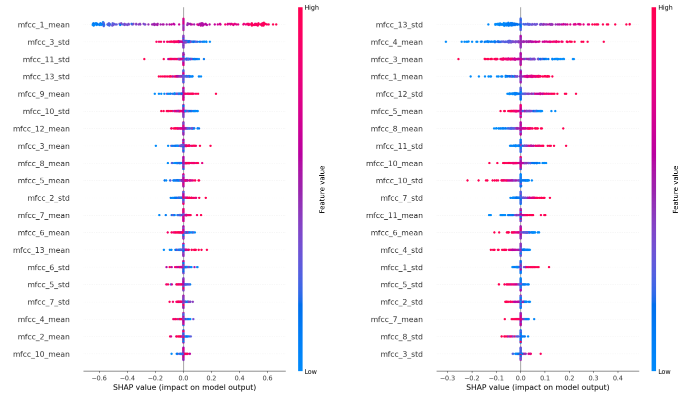
5) SVM Explainability Results: In the present report, the explainability experiments are carried out primarily on the MFCC-SVM baseline, since its interpretable feature space allows a cleaner comparison across SHAP, LIME, and variance-based methods. This choice is also empirically motivated. On VCDetection10K, the linear SVM already achieves near-perfect performance and even outperforms the spectrogram CNN baselines reported later, which makes it a meaningful object of explanation rather than a purely illustrative toy model. At the same time, because its inputs are hand-crafted acoustic descriptors rather than deep latent features, the resulting explanations are easier to interpret acoustically. Extending the analysis to deep spectrogram and anti-spoofing backbones such as VGG16, ResNet18, and AASIST3 is left for future work.

We apply SHAP, LIME, and variance-based Sobol analysis to the linear SVM classifiers trained on VCDetection10K and TTA in order to identify which MFCC statistics drive the real/fake decision. All three methods operate on the same 26-dimensional scaled feature space (13 MFCC means and 13 standard deviations). Below we present global and local explanations for each method, followed by a cross-method comparison.

SHAP Analysis. We use the model-agnostic `KernelExplainer` from the SHAP library [7] with a background set of 100 training samples to marginalise over absent features. Shapley values are computed for 200 test instances with 100 coalitions per evaluation.

VCDetection10K. The beeswarm plot in Figure 13(a) reveals that `mfcc_1_mean` is by far the most influential feature for classifying an utterance as fake, with individual SHAP values reaching ± 0.6. High values of this coefficient (red dots at positive SHAP) push the prediction towards the fake class, while low values push it firmly towards real. The next most important features—`mfcc_3_std`, `mfcc_11_std`, and `mfcc_13_std`—contribute roughly an order of magnitude less. This concentrated importance structure is consistent with the near-perfect classification accuracy (0.996): voice-conversion artefacts leave a strong, almost univariate cepstral signature.

TTA. On the TTA dataset [Figure 13(b)], feature importance is substantially more diffuse. The top-ranked feature, `mfcc_13_std`, reaches SHAP magnitudes of roughly ± 0.3—half of the VC peak—and is followed closely by `mfcc_4_mean`, `mfcc_3_mean`, and `mfcc_1_mean`. No single MFCC dominates the decision; instead, multiple mean and standard-deviation features share comparable Shapley magnitudes. The broader distribution of importance mirrors the weaker SVM accuracy (0.760) and confirms that Text-to-audio generators produce ambient recordings whose cepstral profiles overlap more substantially with natural audio.



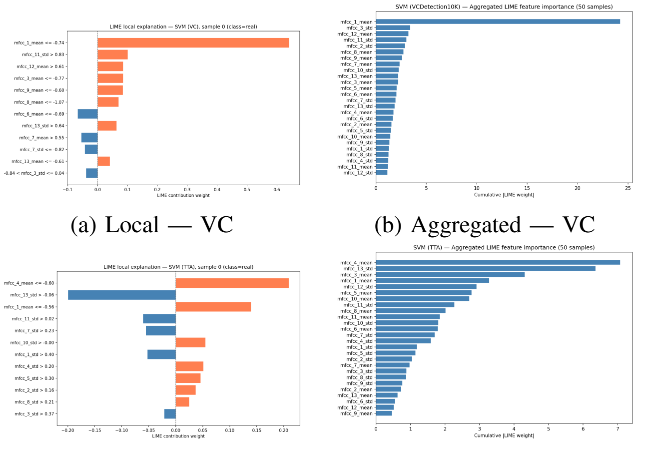
(a) VCDetection10K (b) TTA

Figure 13. SHAP beeswarm plots for the SVM classifier (class = fake). Each dot is one test sample; horizontal position indicates the SHAP value (contribution to the fake-class score), and colour encodes the feature value from low (blue) to high (red). Features are sorted by mean absolute SHAP value. (a) On VCDetection10K, `mfcc_1_mean` dominates. (b) On TTA, importance is spread across many features.

LIME Analysis. A `LimeTabularExplainer` is fit for each test sample using 5 000 perturbed neighbours for individual explanations and 1 000 neighbours for the aggregated analysis. We report the top 12 features per local explanation and aggregate absolute LIME weights over 50 test samples to approximate global importance.

VCDetection10K. The local explanation for a correctly classified real sample (sample 0, $P(\text{fake}) = 0.000$) is shown in Figure 14(a). The surrogate model assigns a dominant weight of $+0.64$ to the condition `mfcc_1_mean` ≤ -0.74, which strongly pushes the prediction towards the real class. The next features—`mfcc_11_std` (0.10), `mfcc_12_mean` (0.09), and `mfcc_3_mean` (0.09)—contribute an order of magnitude less. A few features push weakly in the opposite direction (blue bars), notably `mfcc_6_mean` and `mfcc_7_mean`. The aggregated ranking over 50 samples [Figure 14(b)] preserves this ordering: the cumulative |LIME weight| of `mfcc_1_mean` (≈ 23) exceeds the combined contribution of the next five features ($\approx 7+6+5+4+4$), confirming that VC artefacts are captured primarily by a single cepstral statistic.

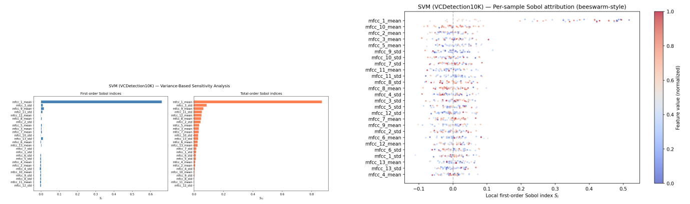
TTA. For TTA, the local LIME explanation of a misclassified fake sample (sample 0, $P(\text{fake}) = 0.346$) assigns the largest weights to `mfcc_4_mean` ($+0.21$) and `mfcc_13_std` (-0.20), the latter pushing against the predicted class [Figure 14(c)]. The importance of `mfcc_1_mean` drops to 0.14, and the overall weight profile is markedly flatter than in the VC case, with many features contributing bidirectionally. The aggregated ranking over 50 TTA samples [Figure 14(d)] confirms a smooth, gradual decay of feature importance with no dominant feature: the top three features (`mfcc_4_mean`,



(a) Local — VC (b) Aggregated — VC

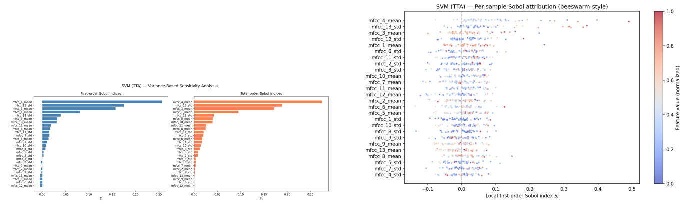
(c) Local — TTA (d) Aggregated — TTA

Figure 14. LIME explanations for the SVM classifier. (a, c) Local bar plots for a single test sample: orange bars push toward the predicted class, blue bars push against it. (b, d) Aggregated cumulative |LIME weight| over 50 test samples, providing a global feature ranking. On VCDetection10K, `mfcc_1_mean` overwhelmingly dominates both locally and globally; on TTA, importance is distributed across several features.



(a) Global S_i vs. S_i^T (b) Per-sample local S_i

Figure 15. Variance-based sensitivity analysis for the SVM on VCDetection10K. (a) Global first-order (blue) and total-order (coral) Sobol indices: `mfcc_1_mean` alone explains ∼55 % of the output variance (S_1) and is involved in ∼85 % when interactions are included (S_1^T). (b) Per-sample local first-order Sobol indices for 50 test instances (dots coloured by normalised feature value): the dominance of `mfcc_1_mean` is consistent across samples.



(a) Global S_i vs. S_i^T (b) Per-sample local S_i

Figure 16. Variance-based sensitivity analysis for the SVM on TTA. (a) Global Sobol indices: importance is distributed across `mfcc_4_mean`, `mfcc_13_std`, `mfcc_1_mean`, and `mfcc_12_std`, with larger $S_i^T - S_i$ gaps indicating stronger interaction effects than in the VC setting. (b) Per-sample local first-order Sobol indices for 50 test instances: greater sample-to-sample variability reflects the noisier TTA decision boundary.

`mfcc_13_std`, `mfcc_1_mean`) each accumulate roughly 4–5.5 units of cumulative weight, in contrast to the VC case where `mfcc_1_mean` alone accumulates ∼23.

Variance-Based (Sobol) Analysis. We compute first-order (S_i) and total-order (S_i^T) Sobol sensitivity indices using 4 096 Saltelli samples drawn within the range of the background distribution. Per-sample local Sobol indices are also computed for 50 test instances to assess sample-level variability.

VCDetection10K. The global Sobol analysis [Figure 15(a)] confirms the dominance of `mfcc_1_mean`: its first-order index is $S_1 \approx 0.55$, meaning that this single feature explains more than half of the output variance of the spoofing score. Its total-order index rises to $S_1^T \approx 0.85$, and the gap $S_1^T - S_1 \approx 0.30$ indicates that `mfcc_1_mean` also participates in interaction effects with other features. By contrast, all other features have near-zero first-order indices and only modest total-order indices (led by `mfcc_3_std` and `mfcc_12_mean`), meaning they influence the classifier primarily through interactions rather than individually. The local beeswarm plot [Figure 15(b)] reveals that `mfcc_1_mean` attains per-sample S_i values as high as 0.5, while most other features remain clustered near zero with occasional outliers. This confirms that the global picture is not driven by a few extreme cases but rather reflects a consistent pattern across the test population.

TTA. The same analysis applied to TTA [Figure 16(a)] yields a markedly different picture. The leading feature, `mfcc_4_mean`, has a first-order index of $S_i \approx 0.25$—less than half of the VC peak—and a total-order index of $S_i^T \approx 0.35$. It is followed by `mfcc_13_std` ($S_i \approx 0.10$, $S_i^T \approx 0.25$), `mfcc_1_mean` ($S_i \approx 0.08$, $S_i^T \approx 0.22$), and `mfcc_12_std` ($S_i \approx 0.06$, $S_i^T \approx 0.18$). The importance is thus spread across many features, and the gap $S_i^T - S_i$ is substantial for several of them (e.g. ∼0.14 for `mfcc_1_mean`, ∼0.15 for `mfcc_13_std`), indicating that interaction effects play a larger role in the TTA setting than in VC. The local beeswarm [Figure 16(b)] confirms this: `mfcc_4_mean` reaches per-sample S_i values up to ∼0.5, but many other features also show non-negligible spread, and the sample-to-sample variability is considerably larger than in the VC case, consistent with the noisier decision boundary and the lower classification accuracy.

Cross-Method Comparison. Table IX summarises the top-5 features identified by each method on both datasets.

On VCDetection10K, all three methods agree unanimously on `mfcc_1_mean` as the dominant feature, and the top-5 lists overlap almost entirely (4 out of 5 features shared). This strong consensus is expected: the near-perfect accuracy indicates that the decision boundary is simple and low-dimensional, so additive (SHAP, LIME) and variance-based (Sobol) decompositions recover essentially the same structure. The Sobol analysis further confirms that interactions are modest ($S_i^T - S_i \leq 0.30$ for the top feature), validating the use of additive surrogates.

On TTA, the three methods still converge on the same small set of features—`mfcc_13_std`, `mfcc_4_mean`, and `mfcc_1_mean`—but differ in their relative ranking. This mild disagreement is expected: when no single feature dominates, the methods' different inductive biases (marginal expectation for SHAP, local linear surrogates for LIME, variance decomposition for Sobol) lead to small rank permutations.

Table IX
TOP-5 MOST IMPORTANT FEATURES ACCORDING TO EACH EXPLAINABILITY METHOD. SHAP AND SOBOL RANKS ARE BASED ON MEAN ABSOLUTE VALUES; LIME RANKS ARE BASED ON CUMULATIVE ABSOLUTE WEIGHTS OVER 50 SAMPLES.

Rank	SHAP	LIME	Sobol (S_i^T)
		VCDetection10K	
1	mfcc_1_mean	mfcc_1_mean	mfcc_1_mean
2	mfcc_3_std	mfcc_3_std	mfcc_3_std
3	mfcc_11_std	mfcc_12_mean	mfcc_12_mean
4	mfcc_13_std	mfcc_11_std	mfcc_3_mean
5	mfcc_9_mean	mfcc_9_mean	mfcc_13_std
		TTA	
1	mfcc_13_std	mfcc_4_mean	mfcc_4_mean
2	mfcc_4_mean	mfcc_13_std	mfcc_13_std
3	mfcc_3_mean	mfcc_1_mean	mfcc_1_mean
4	mfcc_1_mean	mfcc_5_mean	mfcc_12_std
5	mfcc_12_std	mfcc_12_std	mfcc_3_mean

Importantly, the qualitative story is consistent across methods: TTA detection relies on a broader coalition of MFCC statistics rather than on a single discriminative cue.

Summary. The explainability analysis yields two main findings. First, voice-conversion artefacts in VCDetection10K are captured almost entirely by `mfcc_1_mean`—the mean of the first cepstral coefficient, which is closely related to overall spectral energy and log-loudness. This single feature suffices for near-perfect discrimination, indicating that VC models introduce a systematic energy-level shift that is easily detectable. Second, Text-to-audio artefacts are spectrally more subtle: detection requires a combination of mid-to-high-order MFCC means (`mfcc_4_mean`, `mfcc_3_mean`) and variability statistics (`mfcc_13_std`, `mfcc_12_std`), none of which individually provides strong discriminative power. This disparity explains the large performance gap between the two tasks and suggests that improving TTA detection will require richer feature representations or non-linear modelling.

IV. CRITICAL ANALYSIS AND CONCLUSION

A. Critical Analysis

Our results support a clear and, in our view, important conclusion: deepfake-audio detection is strongly *generator-dependent*. The most consistent pattern across the report is not the dominance of one detector over all others, but the large gap between the VC and TTA settings. Voice-conversion deepfakes are comparatively easy to detect, even for simple models. Text-to-audio deepfakes, by contrast, are harder, less stable across detectors, and more sensitive to representation choice. This difference appears repeatedly: in the near-perfect VC performance of the SVM and AASIST3, in the much weaker TTA performance of the SVM, in the performance drop of AASIST3 from VC to TTA, and in the poor cross-domain transfer of the spectrogram CNNs.

This observation also reframes how the generation results should be interpreted. On the generation side, DDDM-VC and Diff-HierVC are closely matched under our automatic VC metrics, while AudioLDM clearly outperforms AudioLDM 2 under our corrected Clotho-based TTA protocol. However, the downstream detection results suggest that generator quality alone is not the whole story. Detectability depends not only on how realistic the generated output appears under objective generation metrics, but also on what residual artifacts remain after synthesis and how easily those artifacts align with the detector representation. In other words, *naturalness* and *detectability* are related, but not identical.

The explainability analysis reinforces this interpretation. On VCDetection10K, SHAP, LIME, and Sobol attribution all converge to the same picture: a very small set of cepstral statistics, especially `mfcc_1_mean`, dominates the decision. This is consistent with a simple and highly structured detection problem. On TTA, by contrast, importance is more distributed, interaction effects are stronger, and no single feature provides dominant separation. This suggests that TTA-generated audio is not merely “harder” in a vague sense, but harder because the discriminative evidence is weaker, more diffuse, and less aligned with simple handcrafted acoustic summaries. That is precisely the kind of phenomenon for which explainability is useful: it turns a performance gap into a more interpretable statement about representation structure.

At the same time, the report has important limitations. First, the final benchmark covers VC and TTA, but not TTS at full scale. Although we reviewed TTS models and partially implemented the branch, the final study does not yet provide controlled generation, detection, and explainability results for that family. Second, the explainability analysis is concentrated on the MFCC-SVM baseline. This was a justified design choice because the SVM is both strong on VC and directly interpretable, but it also means that we do not yet know whether deep spectrogram CNNs or AASIST3 rely on analogous cues, or whether they exploit qualitatively different artifact structures. Third, some comparisons remain setup-dependent. In particular, our TTA conclusions are tied to a Clotho-based protocol rather than the AudioCaps-based settings emphasized in the original AudioLDM papers [3], [4]. The report should therefore be read as a controlled comparative study under our own pipeline, not as a strict reproduction of the source benchmarks.

A further limitation concerns robustness and generalization. The CNN transfer experiments show that strong in-domain performance does not imply cross-domain reliability, which is consistent with broader warnings from the deepfake-detection literature [13]. Even AASIST3, despite being much stronger overall, still exhibits a substantial drop from VC to TTA. This means that the field is still far from a universal audio-deepfake detector. Current systems remain effective mainly when the training and test distributions are sufficiently aligned.

Finally, the contribution of this project is not only empirical but also organizational. By connecting generation, detection, and explainability in a single reproducible pipeline, the report offers a more integrated perspective than a detector-only benchmark would. The released datasets and code are part of that contribution: they make it easier to revisit the current results, extend the benchmark to TTS or multimodality, and test whether future detectors close the generator-dependent gap observed here.

Overall, the strongest critical takeaway is that audio deepfakes should not be treated as a single monolithic category. The difficulty of the problem depends substantially on *which generator family produced the fake*, *which representation the detector uses*, and *which cues remain explainably accessible*. Any claim of robust deepfake-audio detection should therefore be qualified by generator family, domain, and generalization regime.

B. Conclusion

This report studied deepfake audio as a pipeline problem linking generation, detection, and explainability. We constructed reproducible synthetic-audio datasets from two generator families—voice conversion and text-to-audio—using DDDM-VC, Diff-HierVC, AudioLDM, and AudioLDM 2 [1]–[4]. We then evaluated three detector families with increasing complexity: an MFCC-SVM baseline, spectrogram CNNs, and AASIST3 [6], [21], [22]. Finally, we used SHAP, LIME, and Sobol attribution to interpret the detection behavior of the SVM [7]–[9].

The main conclusion is that *voice conversion leaves stronger detectable traces than text-to-audio*. This finding is stable across the report and is more important than any single model ranking. On the generation side, the VC models are closely matched, while AudioLDM outperforms AudioLDM 2 under our corrected Clotho-based evaluation. On the detection side, however, a much stronger pattern emerges: VC deepfakes are consistently far easier to detect than TTA deepfakes. AASIST3 is near-perfect on VCDetection10K but degrades substantially on TTA; the MFCC-SVM shows the same qualitative trend; and spectrogram CNNs generalize poorly across domains. The explainability analysis provides a concrete interpretation of this gap: VC detection is driven by a small number of strong cepstral cues, whereas TTA detection depends on weaker and more distributed statistics.

These findings have two broader implications. First, audio deepfake detection should not be framed as a single homogeneous task. Different generator families induce different artifact structures and therefore different detection regimes. Second, explainability is not only useful for model transparency, but also for understanding *why* some synthetic-audio families are easier to detect than others. In our case, it reveals that the VC/TTA gap is tied to the concentration, strength, and interaction structure of the discriminative cues.

The project also has clear limitations. The TTS branch was reviewed and partially implemented but not completed at benchmark scale, and the explainability study was focused on the interpretable SVM rather than extended to the deep backbones. These are meaningful gaps, but they do not undermine the central result. Instead, they sharpen the most useful lesson of the report: progress in deepfake-audio detection should be evaluated not only by average benchmark accuracy, but by robustness across generator families, domains, and representations.

In summary, our study suggests that current detectors can be extremely effective in favorable in-domain settings, especially for voice conversion, but that robustness remains the key unresolved challenge. The released datasets and code are intended to support follow-up work on broader generator coverage, deeper explainability, and stronger cross-domain detection.

C. Future Work

Several promising directions remain beyond the scope of this project. First, on the *generation* side, future work should expand both the diversity of models and the diversity of audio types considered. In particular, it would be valuable to include stronger and more varied TTS, VC, and TTA systems, and to move beyond speech-only settings toward general audio, including music, singing, and sound effects [42], [43].

Second, on the *explainability* side, our work mainly relies on post-hoc analysis. A natural extension would be to explore *ante-hoc* or intrinsically interpretable detectors, where the prediction is explicitly mediated by human-interpretable acoustic concepts rather than explained only after training [44], [45]. This could provide more trustworthy and stable explanations, especially in ambiguous cases [15].

Finally, an important longer-term direction is *multimodality*. Real-world deepfakes often combine audio with video, text, or metadata, so extending the project toward joint audio-video-text detection would make the framework more realistic and more robust [46]–[48]. Altogether, these directions would broaden the scope of the project while improving both realism and interpretability.

V. Contributions

This project involved all team members, but the contribution balance was not uniform in practice. The majority of the implementation, integration, and report-writing effort was carried by Amine Maazizi, while the other members contributed more specific components of the project. The allocation below reflects the actual work distribution as accurately as possible.

A. Member-Wise Contributions

- **Amine Maazizi:** Led the project overall. Conducted the main literature study for generation, implemented and evaluated the VC and TTA generation pipelines, and built the associated dataset-construction and evaluation pipelines. He wrote the generation-related literature review, methodology, and results sections for both VC and TTA, and also wrote major parts of the report including the abstract, introduction, conclusion, and appendices. In addition, he took the initiative to write the TTS literature and methodology subsections, the CNN-detection bibliography and methodology subsection, completed missing parts of the bibliography across the report, reviewed and helped integrate results produced by other members, and handled a large share of the overall codebase integration, project coordination, and final report consolidation.

- **Adam Miraoui:** Studied and implemented the SVM-based detection and explainability pipeline. Wrote the SVM methodology subsection and the explainability methodology and results subsections, including the SHAP, LIME, and variance-based analyses.
- **Ayoub Boufous:** Studied and implemented the CNN-based detection baselines, including the spectrogram-based VGG16 and ResNet18 experiments. Wrote the corresponding CNN results subsection.
- **Yuheng Zhang:** Studied and implemented the advanced AASIST3-based detection framework. Contributed the AASIST3 bibliography, methodology, and results subsections.
- **Jin Zhuo:** Studied the TTS branch and implemented the Glow-TTS part of the exploratory TTS pipeline. Contributed to the preliminary TTS exploration, although this branch was not completed at benchmark scale.
- **Chenxi Yang:** Explored preliminary XAI directions for TTA, including prompt-variation based experiments. This work remained exploratory and was not fully integrated into the final benchmark.

B. Task Distribution Summary

At a high level, the project was organized around generation, detection, explainability, and report integration. The finalized experimental core relied mainly on the VC and TTA branches for generation, three detector families for downstream comparison, and post-hoc explainability centered on the SVM baseline. In practice, however, the workload was highly skewed. The largest share of implementation, evaluation, integration, and writing was carried by Amine Maazizi. Substantive secondary contributions were made by Adam Miraoui on SVM/XAI, Ayoub Boufous on CNN baselines, and Yuheng Zhang on AASIST3. The TTS and exploratory TTA-XAI directions remained more limited, with partial contributions from Jin Zhuo and Chenxi Yang respectively.

The project outputs include not only the report itself, but also the associated codebase and dataset resources released by the team through the project repository and dataset hosting platform.

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Appendix A
DDPM Background

Denoising Diffusion Probabilistic Models (DDPMs) define a fixed forward Markov chain that gradually corrupts a data sample x_0 into Gaussian noise, together with a learned reverse chain that reconstructs the sample from noise [49]. Using the standard variance schedule $\{\beta_t\}_{t=1}^{T}$, the forward process is

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\, x_{t-1},\, \beta_t I\Big),$$

which implies the closed-form marginal

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\, (1-\bar{\alpha}_t)I\big), \qquad \bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s).$$

Generation is defined by the reverse model

$$p_\theta(x_{0:T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t), \qquad p(x_T) = \mathcal{N}(0, I),$$

with Gaussian reverse transitions

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\, \sigma_t^2 I\big).$$

With the noise-prediction parameterization, training reduces to the simplified objective

$$\mathcal{L}_{\mathrm{DDPM}} = \mathbb{E}_{x_0,\epsilon,t}\left[\left\|\epsilon - \epsilon_\theta\Big(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon,\, t\Big)\right\|_2^2\right], \\ \epsilon \sim \mathcal{N}(0, I).$$

Hence, DDPM learns to predict the Gaussian perturbation added at each timestep, and sampling consists of initializing $x_T \sim \mathcal{N}(0, I)$ and iteratively applying the learned reverse denoising transitions [49].

Appendix B
DDIM Background

Denoising Diffusion Implicit Models (DDIMs) keep the same denoising training objective as DDPMs, but reinterpret sampling through a family of non-Markovian processes with the same marginals [50]. Using the DDPM cumulative notation, one first forms the denoised estimate

$$\hat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

The generalized DDIM sampling update can then be written as

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2}\,\epsilon_\theta(x_t, t) + \sigma_t z, \\ z \sim \mathcal{N}(0, I).$$

The parameter σ_t controls the stochasticity of the reverse trajectory. When $\sigma_t = 0$, the update becomes deterministic, which yields the implicit DDIM sampler:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\,\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\,\epsilon_\theta(x_t, t).$$

Therefore, DDIM does not introduce a new training loss; rather, it provides an alternative inference procedure that can use fewer timesteps while often preserving good sample quality. This is why DDIM is mainly relevant when discussing accelerated sampling and inference-time trade-offs in diffusion-based generation [50].

Appendix C
Detailed Dataset Construction

A. Voice-Conversion Dataset Generation

For the voice-conversion branch, we started from the VCTK corpus and extracted a reproducible subset of short real utterances suitable for conversion-based deepfake generation [51]. In our pipeline, the raw clips were standardized to a common format, filtered by duration, and stored together with transcripts and speaker identifiers in a metadata manifest. A key design choice was to enforce speaker-disjoint train/validation/test splits, so that speaker identity did not leak across partitions. In the configuration used for the main experiments, we targeted a balanced real subset and retained full provenance information for each clip, including split, speaker ID, path, duration, sample rate, and transcript.

We then generated deterministic source–target conversion pairs for the diffusion VC models. For each selected source utterance, the pipeline sampled a target speaker reference from the same data split, with different-speaker pairing enforced whenever possible. Separate manifests were produced for DDDM-VC and Diff-HierVC, and the pair construction logic was designed to keep source usage controlled across models in order to reduce generator-specific bias. Since this stage was inference-only and the VCTK clips were relatively short, generation was run locally on a laptop GPU (RTX 4070), which was sufficient in practice without requiring large-memory multi-GPU infrastructure. The resulting converted files were tracked through completion manifests and then merged with the real clips to build the final detection datasets. We exported two variants: a binary dataset with *real* and *fake* labels, and a multiclass dataset that preserved the generator identity of each fake sample. In both cases, metadata preserved the link between each detection item and its original source clip, conversion pair, model name, and audio file.

B. Text-to-audio Dataset Generation

The Text-to-audio branch followed the same general philosophy, but with caption-conditioned generation rather than speaker-conditioned conversion. We first validated the generation and evaluation stack on a small pilot subset, then built the final large-scale detection dataset from a reproducible caption-audio subset derived from Clotho [52]. Ground-truth clips and captions were standardized through a manifest-based preprocessing stage, after which a single train/validation/test partition was fixed and propagated through all subsequent stages. This ensured that the split structure remained consistent from real data extraction to synthetic generation and final detection-dataset assembly.

Synthetic audio was then generated with two Text-to-audio systems, AudioLDM and AudioLDM 2. For each model, we created a dedicated generation manifest containing the caption, split, deterministic seed, inference hyperparameters, output path, and execution status. Large-scale generation was not run locally: although it remained an inference workload, it was substantially heavier than the VC case and the input material was more variable in duration and acoustic complexity. For that reason, the TTA synthesis jobs were executed on the school cluster using NVIDIA H100 GPUs. The pipeline was also designed to be resumable, so partially completed generations could be continued without restarting the entire run. After synthesis, the generated audio was merged with the real subset to construct the final balanced detection datasets. As in the VC branch, we exported both a binary real-vs.-fake version and a generator-aware multiclass version, while keeping metadata fields such as caption, source sample ID, model name, audio path, duration, and sample rate for downstream evaluation and analysis.

APPENDIX D
SPLIT-WISE GENERATION RESULTS

A. Voice-Conversion

Table X reports the split-wise VC results underlying the aggregated comparison in the main text. CER, WER, and EER are reported in %, while SECS is reported on $[0, 1]$. These results are included for transparency and to illustrate the variability of the relative ranking across train, validation, and test subsets.

Table X
SPLIT-WISE VC RESULTS UNDER OUR EVALUATION SETUP. LOWER IS BETTER FOR CER, WER, AND EER; HIGHER IS BETTER FOR SECS.

Model	Split	CER↓	WER↓	EER↓	SECS↑
DDDM-VC	Test	74.09	90.60	38.25	0.6728
DDDM-VC	Train	108.09	131.42	47.73	0.5700
DDDM-VC	Val	72.09	84.41	48.23	0.6731
Diff-HierVC	Test	109.05	133.63	51.53	0.6465
Diff-HierVC	Train	110.37	133.96	50.03	0.5646
Diff-HierVC	Val	41.95	51.76	36.48	0.7411

B. Text-to-audio

Table XI reports the split-wise Text-to-audio results underlying the aggregated comparison in the main text. Higher is better for CLAP, while lower is better for FAD and KL. These results are included for transparency and to show that, under our corrected evaluation pipeline, the same ranking is observed across train, validation, and test subsets.

Table XI
SPLIT-WISE TTA RESULTS UNDER OUR CORRECTED CLOTHO V2.1 EVALUATION SETUP. HIGHER IS BETTER FOR CLAP; LOWER IS BETTER FOR FAD AND KL.

Model	Split	CLAP↑	FAD↓	KL↓
AudioLDM	Train	0.327	3.084	0.616
AudioLDM	Test	0.303	4.669	0.696
AudioLDM	Val	0.317	4.962	0.976
AudioLDM 2	Train	0.261	4.549	1.328
AudioLDM 2	Test	0.262	5.700	1.517
AudioLDM 2	Val	0.286	4.425	1.204